

ADVISORY CIRCULAR

SUBJECT: Pilot Training and Certification for Manned Powered-Lift (VTOL) Operations.	DATE: 2026-04-16	AC NUMBER: AC 100-02	VERSION: 1.0
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NOTE: THIS ADVISORY CIRCULAR IS PUBLISHED TO PROVIDE REGULATORY INFORMATION AND DESCRIBE ACCEPTABLE MEANS OF COMPLIANCE WITH THE GENERAL AUTHORITY OF CIVIL AVIATION REGULATIONS (GACAR).

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CHAPTER 1. INTRODUCTION

1.1 Purpose of This Advisory Circular (AC). This AC provides guidance for applicants, pilots, evaluators, and instructors who seek to add an initial powered-lift category rating and specific type rating to their existing pilot certificate in accordance with GACAR Part 100. Pilots and certificate holders (CH) are encouraged to use this information as best practice methods for developing training programs scaled to specific powered-lift they operate. The contents of this document do not have the force and effect of law and are not meant to bind the public in any way, and the document is intended only to provide information to the public regarding existing requirements under the GACARs.

1.2 Audience. The primary audience for this AC is persons seeking to obtain an initial powered-lift category rating, a specific type rating, and/or an instrument-powered-lift rating to an existing pilot or flight instructor certificate (e.g., applicants, pilots, and flight instructors) and entities seeking to provide training to such persons (e.g., GACAR Part 135 operators, GACAR Part 141 pilot schools, and GACAR Part 142 training centers). Designated Pilot Examiners (DPE), aviation safety inspectors (ASI), and other persons who conduct certification functions for pilots should familiarize themselves with this AC as well.

1.3 Where You Can Find FAA Acs related to this GACA AC. You can find related FAA ACs on the Federal Aviation Administration's (FAA) website at https://www.faa.gov/regulations_policies/advisory_circulars and the Dynamic Regulatory System (DRS) at <https://drs.faa.gov>.

1.4 Effective Date. This AC is effective as of its date of publication.

1.5 Related GACAR Parts. The following are available at <https://www.gaca.gov.sa>:

- GACAR Part 61, Certification: Pilots, Flight Instructors, and Ground Instructors.
- GACAR Part 91 Subpart D, Special Flight Operations.
- GACAR Part 135, Operations: Commercial Air Operators Operating Other Than Transport Category Aircraft or Commuter Category Airplanes
- GACAR Part 141, Pilot Schools.
- GACAR Part 142, Training Centers.

1.6 Related Airman Certification Standards (ACS) (current editions). The following should be read in conjunction with this AC and are available at https://www.faa.gov/training_testing/testing/acs:

- FAA-S-ACS-2, Commercial Pilot for Powered-Lift Category Airman Certification Standards.

- FAA-S-ACS-3, Instrument Rating—Powered-Lift Airman Certification Standards.
- FAA-S-ACS-13, Private Pilot for Powered-Lift Category Airman Certification Standards.
- FAA-S-ACS-17, Airline Transport Pilot and Type Rating for Powered-Lift Category Airman Certification Standards.
- FAA-S-ACS-27, Flight Instructor for Powered-Lift Category Airman Certification Standards.
- FAA-S-ACS-28, Flight Instructor—Instrument Rating Powered-Lift Airman Certification Standards.

1.7 Definitions. The following definitions are only for the purposes of this AC and GACAR Part 100.

1.7.1 *Alternate Definition of Cross-Country Time (GACAR § 100.201)*. Notwithstanding the cross-country time definition in GACAR § 1.1, a person may log flight time in a powered-lift as cross-country time provided the time was acquired during a flight:

1. That includes a point of landing that was at least a straight-line distance of more than 25 nautical miles (NM) from the original point of departure; and
2. That involves the use of dead reckoning (DR), pilotage, electronic navigation aids, radio aids, or other navigation systems to navigate to the landing point.

1.7.2 *Aviation Safety Inspector (ASI) (GACAR § 100.103)*. A pilot employed by the GACA to conduct operations of a powered-lift for the purpose of establishing a type rating in that specific powered-lift under GACAR Part 21.

1.7.3 *Evaluator*. For the purposes of this AC, an evaluator is considered:

- An ASI;
- A pilot examiner (other than administrative pilot examiners);
- An initial check pilot;
- A Training Center Evaluator (TCE);
- An Aircrew Program Designee (APD); or
- A chief instructor, an assistant chief instructor, or a check instructor of an approved GACAR Part 141 pilot school.

1.7.4 *CAA test pilot (GACAR § 100.103)*. A pilot employed by the GACA, or otherwise authorized by the President, to act on behalf of the GACA to conduct operations of a powered-

lift for the purpose of examination or inspection of a type design for which an application for type certification or amended type certificate has been submitted under GACAR Part 21.

1.7.5 Instructor Pilot (GACAR § 100.103). A pilot employed or used by a manufacturer of a powered-lift to conduct operations of the powered-lift for the purpose of developing a proposed training curriculum and providing crew training. This term is not the same as an authorized instructor.

1.7.6 Manufacturer (GACAR § 100.103). Any person who holds, or is an applicant for, a type or production certificate for an aircraft. An amateur builder or a builder of a kit aircraft under GACAR § 21.173, or the holder of a restricted category type certificate (TC) is not considered a manufacturer for the purposes of GACAR Part 21.

1.7.7 Test Pilot (GACAR § 100.103). A pilot employed or used by a manufacturer of a powered-lift to conduct operations of the powered-lift for the purpose of research and development (R&D) and showing compliance with the regulations.

1.8 Background.

1.8.1 As a result of the rapidly changing technology and newly designed aircraft intending to fly in KSA airspace, the GACA created regulations to account for the new category of aircraft. A portion of the regulatory changes include creating pilot training and certification requirements for a new aircraft category. GACAR Part 100 provides regulatory requirements for powered-lift pilot certification and operations. Prior to this regulation, the GACA did not have a feasible means to implement training and certification of pilots for a powered-lift category rating due to a lack of certificated aircraft and certified personnel to provide instruction. This new regulation establishes requirements for qualifying pilots in powered-lift by utilizing new and existing pathways for pilot training, checking, and testing in powered-lift under GACAR Parts 61, 91, 135, 141, 142, and 100. Additionally, GACAR Part 100 provides for alternate pathways for training and certification of pilots, as there is currently limited means to obtain experience in a powered-lift.

1.8.2 Currently, civilian pilots are unable to satisfy many of the aeronautical experience requirements in GACAR Part 61 to obtain a Commercial Pilot Certificate with a powered-lift category rating and/or an instrument-powered-lift rating because there are no certificated powered-lift in civil operations in which they can obtain the necessary flight time. At the time of publication of GACAR Part 100 there are currently no type-certificated civilian powered-lift; however, there are many manufacturers starting the production process.

1.8.3 When powered-lift are introduced to civil aviation, pilots will initially be unable to satisfy several of the aeronautical experience requirements for a Commercial Pilot Certificate with a powered-lift category rating and an instrument-powered-lift rating (such as pilot-in-command (PIC) flight time in powered-lift and cross-country experience in powered-lift) due to lack of qualified civilian pilots and instructors in powered-lift. Further, the PIC logging requirements of

GACAR § 61.13(d) currently present obstacles for a pilot who is not rated in a powered-lift to log PIC flight time in a powered-lift. Additionally, in light of the range limitations of current type-certification projects for powered-lift, and several planned projects, many powered-lift coming to market are not capable of completing the prescribed cross-country aeronautical experience requirements for the Commercial Pilot Certificate and instrument rating. GACAR Part 100 facilitates pathways for pilots to obtain powered-lift ratings through alternate aeronautical experience and expanded logging provisions.

1.9 Reserved.

CHAPTER 2. POWERED-LIFT PILOT CERTIFICATION—GENERAL PROVISIONS

2.1 General. GACAR Part 100 maintains the regulatory framework of powered-lift as an independent aircraft category without any associated classes. The GACA expects that, initially, most pilots seeking to add a powered-lift category rating to their pilot certificate will already hold another category rating (i.e., airplane or rotorcraft) on their certificate. In fact, as subsequently discussed, to utilize most of the alternate aeronautical experience requirements and logging provisions, pilots are required by GACAR § 100.215 to hold certain certificates and ratings. Permanent changes to GACAR Part 61 as well as temporary alternate experience requirements in GACAR Part 100, work in tandem to provide methods for certification for all powered-lift pilots. The next several chapters outline the temporary and permanent changes that affect pilot training, checking, and testing in powered-lift.

2.2 Type Rating Requirements. To address the expected significant operational differences between each powered-lift, GACAR § 61.85(a) requires each person who acts as pilot in command (PIC) of a powered-lift to hold the associated type rating. Requiring pilots to hold type ratings for powered-lift establishes a higher level of safety by ensuring persons receive adequate training and are tested on the unique design and operating characteristics of each powered-lift. The GACA will identify the unique characteristics of each powered-lift that require special training. Subsequently, the this required special training should be utilized in conjunction with the appropriate powered-lift Airman Certification Standards (ACS) to conduct training and practical tests for a type rating, ensuring that an applicant is knowledgeable and capable of safely operating the unique powered-lift type.

2.3 Applicability of the Type Rating Requirement to Military Pilots.

2.3.1 Military pilots with powered-lift qualifications may apply for Commercial Pilot Certificates with powered-lift category ratings and instrument-powered-lift ratings, and flight instructor certificates with powered-lift category ratings and instrument-powered-lift ratings, pursuant to the existing requirements in GACAR § 61.93, based on their military pilot qualifications.

2.3.2 However, to act as PIC of a civilian powered-lift, these military pilots are required to obtain a type rating on their pilot certificate for the applicable type of civilian powered-lift, as required by GACAR § 61.85(a). Those military instructors who have obtained or plan to obtain a civil flight instructor certificate with a powered-lift category rating and instrument-powered-lift rating through military competency may conduct flight training in a powered-lift only after obtaining a type rating on their pilot certificate for the powered-lift in which they want to conduct flight training.

CHAPTER 3. ALTERNATE CROSS-COUNTRY REQUIREMENTS

3.1 Alternate Means to Log Cross-Country Time in Powered-Lift.

3.1.1 Due to the expected disparity between powered-lift range capabilities, GACAR Part 100 includes an alternate definition of cross-country time that reduces the general distance to log cross-country time in a powered-lift from more than 50 nautical miles (NM) to more than 25 NM. Specifically, to log flight time in a powered-lift as cross-country time, GACAR § 100.201 requires a point of landing that is at least a straight-line distance of more than 25 NM from the original point of departure and involves the use of dead reckoning (DR), pilotage, electronic navigation aids, radio aids, or other navigation systems to navigate to the landing point.

3.1.2 For applicants not qualified to act as pilot in command (PIC) in a powered-lift, GACAR § 100.213 requires all persons conducting cross-country flights to receive an endorsement from an authorized instructor or instructor pilot. Since applicants are not qualified to be PIC in a powered-lift, they will need an endorsement from the authorized instructor or instructor pilot stating that the applicant meets the alternate cross-country requirements. The applicant's logbook must demonstrate to the evaluator that they have met the alternate cross-country requirements of GACAR Part 100 in lieu of the standard GACAR Part 61 regulatory requirements. Appendix A of this AC provides an example of each endorsement.

3.1.3 In addition to the cross-country definition, GACAR Part 100 provides relief from specific cross-country requirements, dependent on the certificate sought, to address possible range capabilities, as previously discussed. Paragraphs 3.2 through 3.4 detail specific relief options for applicants.

3.1.4 To note, a pilot may satisfy either the traditional GACAR Part 61 cross-country pathway or the alternate cross-country requirements set forth by GACAR Part 100, as applicable, to meet the requisite cross-country aeronautical experience.

3.2 Commercial Applicants.

3.2.1 Under Appendix D, paragraph (e)(3) to GACAR Part 61, applicants for a powered-lift rating are required to complete only two cross-country flights, one in daytime conditions and one in nighttime conditions, each including a total straight-line distance of more than 100 NM from the original point of departure. However, as many powered-lift are expected to be unable to comply with existing cross-country requirements, the alternate requirements in GACAR Part 100 reduce the distance required for the cross-country flight to a total straight-line distance of more than 50 NM and add an additional cross-country flight with landings at a minimum of three points as a safety mitigation. GACAR § 100.233(a)(3) requires that, other than the original point of departure, the third cross-country flight must be performed to different points than the first two cross-country flights at the distances specified. The third cross-country flight may be

conducted during day or nighttime conditions. This additional cross-country flight will add to the applicant's overall experience in the specific powered-lift in which the applicant is seeking their initial Commercial Pilot Certificate with powered-lift category rating.

3.2.2 In lieu of complying with Appendix D, paragraph (e)(3) to GACAR Part 61, GACAR § 100.233 allows an applicant seeking a Commercial Pilot Certificate in a powered-lift to complete the following cross-country flights:

1. One 2-hour cross-country flight in a powered-lift in daytime conditions that consists of a total straight-line distance of more than 50 NM from the original point of departure;
2. One 2-hour cross-country flight in a powered-lift in nighttime conditions that consists of a total straight-line distance of more than 50 NM from the original point of departure; and
3. An additional cross-country flight with landings at a minimum of three points, with one segment consisting of a straight-line distance of at least 50 NM from the original point of departure. This flight must include landings at different points than the points selected for the previous two cross-country flights.

3.3 Instrument Applicants. Appendix E, paragraph (c)(2)(ii) to GACAR Part 61 requires completion of one cross-country flight of 250 NM. In lieu of complying with this requirement, GACAR § 100.235 allows an applicant seeking an instrument rating in a powered-lift to complete two cross-country flights, each of which must involve a flight of 100 NM along airways or by directed routing from an air traffic control (ATC) facility. Thus, each cross-country flight with the reduced 100 NM distance must be performed under instrument flight rules (IFR) when a flight plan has been filed with an ATC facility and must involve an instrument approach at each aerodrome and three different kinds of approaches with the use of navigation systems. However, an applicant utilizing the alternate means under GACAR § 100.235 is not required to file a flight plan and perform the cross-country flight under IFR if the powered-lift is not certificated for instrument flight and the applicant holds either:

- An instrument-airplane rating,
- An instrument-helicopter rating, or
- An Airline Transport Pilot (ATP) Certificate with an airplane category rating or rotorcraft category and helicopter class ratings.

3.3.1 In lieu of complying with Appendix E, paragraph (c)(2)(ii) to GACAR Part 61, GACAR § 100.235 allows an applicant seeking an instrument-powered-lift rating to complete the following cross-country flights two cross-country flights conducted with either an authorized instructor or an instructor pilot that involve a flight of 100 NM along airways or by directed routing from an ATC facility, include an instrument approach at each aerodrome, and include three different kinds of approaches with the use of navigation systems.

3.4 Private Pilot Applicants.

3.4.1 GACAR § 100.201 authorizes private pilot applicants to apply the alternate definition of cross-country time when seeking to comply with the cross-country flight training in GACAR § 61.133. In addition, instead of completing one cross-country flight of over 100 NM distance at night, as currently required by Appendix C, paragraph (e) to GACAR Part 61 GACAR § 100.237(a) allows an applicant to complete two cross-country flights at night, each of which involves a flight of over 50 NM total distance.

3.4.2 Additionally, in place of the solo cross-country flight that is currently required by Appendix C, paragraph (e) to GACAR Part 61 (i.e., 150 NM total distance with one segment of the flight consisting of a straight-line distance of more than 50 NM), GACAR § 100.237(b) allows an applicant to complete the following alternate requirements:

1. One solo cross-country flight of 100 NM total distance, with landings at three points, and one segment of the flight being a straight-line distance of more than 25 NM between the takeoff and landing locations; and
2. An additional solo cross-country flight in a powered-lift with landings at a minimum of three points, with one segment consisting of a straight-line distance of at least 50 NM from the original point of departure. This additional solo cross-country flight requires landings at a minimum of three points, with one segment consisting of a straight-line distance of at least 25 NM from the original point of departure. This additional cross-country flight must include different points of landing than the initial cross-country flight.

CHAPTER 4. ALTERNATE REQUIREMENTS FOR PRACTICAL TESTS AND TRAINING IN A POWERED-LIFT

4.1 Required Equipment for Practical Tests in a Powered-Lift.

4.1.1 GACAR § 61.27(c)(2) stipulates the equipment, other than controls, required of an aircraft used on the practical test and allows the use of an aircraft with operating characteristics that preclude the applicant from performing all the tasks for a practical test. In this case, an appropriate limitation is placed on the applicant's certificate or rating. However, GACAR § 100.207(a) permits an applicant for a certificate or rating to use a powered-lift that is precluded from performing all of the tasks required for the practical tests. Applicants will not be required to have the certificate or rating issued with an appropriate limitation, as is currently required by GACAR § 61.27(c)(2)(ii). The purpose of the limitation under GACAR § 61.27(c)(2)(ii) is to ensure the pilot operates only the make and model (M/M) within the category and class of aircraft brought to the test, thereby preventing the pilot from acting as pilot in command (PIC) of an aircraft capable of performing tasks for which the pilot has not been tested. However, in the case of pilots seeking a certificate with associated powered-lift ratings, GACAR § 61.85(a)(4) requires all pilots to receive a PIC type rating for each aircraft type. Therefore, such a limitation is not necessary because GACAR § 61.85(a)(4) requires applicants to take a practical test in each powered-lift they seek to operate to act as PIC. In other words, because all powered-lift will require a type rating, a pilot is already limited to acting as PIC of only those powered-lift for which they hold a type rating (i.e., for which they have passed a practical test).

4.1.2 For example, if type A powered-lift could not perform a stall, then a pilot seeking a type rating in type A would not be tested on stalls and would not receive a limitation on the type rating for type A. If the pilot wished to act as PIC of type B powered-lift, which is capable of performing a stall, the absence of a limitation for the type rating for type A would not present a safety concern because the pilot would need to take a practical test for a type rating in type B powered-lift, pursuant to GACAR § 61.85(a), which would include the previously omitted evaluation on stalls. In other words, the type rating itself contains the limitation contemplated in GACAR § 61.27(c)(2)(ii) for an aircraft not able to perform all tasks in the Airman Certification Standards (ACS).

4.1.3 GACAR § 61.27(b)(1)(i) requires that the completion of a practical test for a certificate or rating includes the performance of the tasks specified in the areas of operation for the airman certificate or rating sought. For powered-lift, these tasks are set forth in the ACS for the appropriate certificate or rating that the applicant is seeking. GACAR § 100.207(b) provides evaluators conducting practical tests for powered-lift the ability to waive any task for which the GACA has provided waiver authority. In order to determine what tasks may be waived, the type certification and evaluation process will be utilized to identify the operational limitations of the powered-lift, which may preclude the conduct of certain tasks listed in the ACS. The GACA (and

based on the FAA Flight Standardization Board Report (FSBR) or equivalent) subsequently will ascertain what tasks in the ACS are not applicable to the specific type of powered-lift. This waiver authority is not at the evaluator's discretion; rather, the waived tasks are set forth on a designee's written authorization from the President specific to each powered-lift type in which they are authorized to conduct practical tests. Specifically, the written authorization will identify the type of powered-lift that the evaluator is authorized to conduct a practical test in, and the specific tasks that the evaluator is authorized to waive for the practical test will be set forth in the limitations section of the written authorization.

4.2 Required Flight Training Tasks. In addition to the required tasks specified in the areas of operation for the airman certificate and rating sought, GACAR § 61.27(a) requires an applicant for a certificate or rating to receive and log flight training on the applicable areas of operation. GACAR § 100.207 authorizes an evaluator to waive a specific task during a practical test because the powered-lift used is precluded of performing a specific task. Therefore, an applicant for a Private Pilot Certificate or Commercial Pilot Certificate with a powered-lift category rating concurrently with a powered-lift type rating is relieved from the requirement to receive flight training on any task specified in an area of operation that is waived on the practical test.

4.2.1 In the case where the GACA identifies that a task is precluded from being accomplished because of factors during aircraft certification, the GACA will update its guidance in order to notify instructors, designees, and the general public on any specific task that is precluded from being trained or tested.

CHAPTER 5. ALTERNATE ELIGIBILITY REQUIREMENTS FOR A PERSON SEEKING AN INITIAL POWERED-LIFT TYPE RATING

5.1 Practical Tests for Powered-Lift Category Rating and Type Rating. If an applicant applies to obtain an initial powered-lift type rating concurrently with a powered-lift category rating and instrument-powered-lift rating, that applicant is required to take three practical tests concurrently under GACAR § 61.83(d)(1). However, GACAR Part 100 provides temporary relief for the initial powered-lift practical tests. GACAR § 100.211(b) provides eligible applicants with an option to obtain a powered-lift type rating without concurrently completing the required instrument rating; in other words, the applicant may take the instrument-powered-lift rating practical test independent from the practical tests for the powered-lift category and type ratings.

5.2 Eligible Applicants. To be eligible to utilize the relief provided in GACAR § 100.211(b) and obtain an initial powered-lift type rating without concurrently holding or concurrently obtaining an instrument-powered-lift rating, a person must be seeking a type rating in a powered-lift capable of performing instrument maneuvers and procedures, pursuant to GACAR § 100.211(a). If the powered-lift in which the type rating is sought is incapable of performing instrument maneuvers and procedures, then the applicant must comply with GACAR § 61.83(e).

Note: Although certain powered-lift may not be instrument flight rules (IFR)-certified initially, they will likely be capable of performing instrument maneuvers and procedures because they will have the equipment necessary to perform instrument maneuvers. In this case, the applicant would be required to comply with either GACAR § 61.83(d) or GACAR § 100.211.

5.3 Practical Tests Required for Persons Seeking an Initial Powered-Lift Type Rating.

5.3.1 Under GACAR § 100.211(b)(1), applicants for an initial powered-lift type rating to be completed concurrently with a powered-lift category rating would not need to concurrently hold or obtain the instrument-powered-lift rating at the same time they obtain their powered-lift type and category ratings. Instead, the applicant applying for an initial powered-lift type rating pursuant to GACAR § 100.211(b) could choose to take the instrument-powered-lift rating practical test independent from the powered-lift type rating practical test and powered-lift category rating practical test. Opting to take the instrument-powered-lift rating practical test separately results in a “visual flight rules (VFR) only” limitation being placed on the type rating. Paragraph 5.6 of this AC describes how to remove the limitation. Commercial applicants will also receive the limitation specified in GACAR § 61.155(b)(1), “The carriage of passengers for hire in powered-lifts on cross-country flights in excess of 50 nautical miles or at night is prohibited.”

5.3.2 The “VFR only” limitation in GACAR § 100.211(b) is a one-time-only option for initial powered-lift type rating applicants that do not wish to be subjected to the burden of taking three practical tests at one time. Except as provided in GACAR § 100.211(c), any additional powered-

lift type ratings require the entire practical test in accordance with FAA-S-ACS-17, Airline Transport Pilot and Type Rating for Powered-Lift Category Airman Certification Standards (ACS).

5.3.3 Additionally, because GACAR § 100.211(b)(2) does not require the type rating test to be taken in actual or simulated instrument conditions, the practical test for the type rating would not require the applicant to perform those areas of operation in the ACS that include instrument maneuvers and procedures in actual or simulated instrument conditions.

5.4 Pilot Privileges With a “VFR Only” Limitation. In accordance with GACAR § 61.89(b), a pilot with a “VFR only” limitation (i.e., the pilot did not concurrently complete the instrument-powered-lift rating practical test and, therefore, does not possess the instrument rating) on their powered-lift type rating is prohibited from operating the powered-lift under IFR even if the powered-lift can perform instrument procedures and maneuvers. The pilot is only permitted to operate the powered-lift under the conditions for which the pilot demonstrated mastery of the powered-lift on the practical test.

5.5 Removal of “VFR Only” Limitation—Private Pilot. Under GACAR § 100.211(b)(6), a pilot holding a Private Pilot Certificate with a powered-lift type rating and category rating with a powered-lift “VFR only” limitation is not required to remove the “VFR only” limitation if the limitation applies to a powered-lift type that is not a large aircraft or turbojet-powered. If a private pilot chooses to remove the “VFR only” limitation, they must comply with the “VFR only” limitation removal requirements set forth by GACAR § 100.211(b)(4), as discussed in paragraph 5.6 of this AC.

5.6 Removal of “VFR Only” Limitation—Other Than Private Pilot. Except for private pilots that operate powered-lift types that are not large aircraft or turbojet-powered, a person who obtains a powered-lift type rating with a “VFR only” limitation must remove the “VFR only” limitation within 2 calendar months from the month in which the person completed the powered-lift type rating practical test, in accordance with GACAR § 100.211(b)(4).

Note: The purpose of the “VFR only” limitation is to permit limited flexibility in the scheduling of the instrument-powered-lift rating practical test at a time separate from the category and type rating practical tests. An applicant is eligible to take the instrument-powered-lift rating practical test at the same time as the powered-lift type rating and powered-lift category rating practical tests. Therefore, the applicant will have obtained the required 3 hours of flight training in preparation for the instrument rating practical test within the 2 calendar months preceding the month of the practical tests, as required by Appendix E, paragraph (f)(2)(i). The applicant should have sufficient time within the 2-calendar-month window for the endorsement to complete all of the practical tests and should not need to accomplish any additional training to be eligible to take the instrument-powered-lift practical test during this time period.

5.7 Practical Test Requirements to Remove the “VFR Only” Limitation.

5.7.1 In accordance with GACAR § 100.211(b)(4), a person with a powered-lift type rating with a “VFR only” limitation may remove the limitation by completing both of the following:

1. Passing an instrument rating practical test in a powered-lift, where the test must be passed in actual or simulated instrument conditions; and
2. Passing a practical test in the powered-lift type for which the “VFR only” limitation applies on the areas of operation listed in GACAR § 61.179(e)(3) that includes performing instrument maneuvers and procedures in actual or simulated instrument conditions.

5.7.2 The required tasks for the instrument-powered-lift rating are found in FAA-S-ACS-3, Instrument Rating—Powered-Lift Airman Certification Standards. The Airline Transport Pilot (ATP) and Type Rating for Powered-Lift Category ACS specifies which areas of operation and tasks an applicant must satisfactorily perform during the type rating practical test to remove the “VFR only” limitation for a powered-lift type.

5.7.3 When an applicant takes concurrent practical tests to remove the “VFR only” limitation from their certificate, they will have to accomplish all required tasks, per GACAR § 100.211(b)(4). However, when a task required for the instrument rating practical test overlaps with a task required for the type rating practical test, GACAR § 100.211(d) allows an applicant to perform the task a single time, provided the person performs the task to the highest standard required for the task.

Example: An applicant with a “VFR only” limitation on their powered-lift type rating applies to take the instrument-powered-lift practical test to remove the limitation. That person would be required to perform a circling approach procedure on both the instrument rating practical test and the type rating practical test.

The Instrument Rating—Powered-Lift ACS requires the applicant to perform the procedures while maintaining an airspeed of ± 10 knots and a desired heading/track of ± 10 degrees. The ATP and Type Rating for Powered-Lift Category ACS also requires the applicant to perform the circling approach procedure, but to more exacting standards (e.g., maintain an airspeed of ± 5 knots and a desired heading/track of ± 5 degrees). If a pilot demonstrates their ability to perform the circling approach procedure to the more exacting standards specified in the ATP and Type Rating for Powered-Lift Category ACS, the pilot would not be required to perform the same task a second time to the less stringent standards specified in the Instrument Rating—Powered-Lift ACS.

5.7.4 In addition, once a commercial applicant successfully completes the instrument practical test, the limitation specified in GACAR § 61.155(b)(1) will also be removed because, consistent with that section, the pilot will have accomplished the requirements listed in GACAR § 61.89 for an instrument rating in the same category of aircraft listed on the person’s Commercial Pilot Certificate.

5.8 Effect of Failure to Remove “VFR Only” Limitation.

5.8.1 If the pilot does not remove the “VFR only” limitation as required by GACAR § 100.211(b)(3), then the powered-lift type rating for which the “VFR only” limitation applies will become invalid for use until the person removes the limitation in accordance with GACAR § 100.206(b)(4). Per GACAR § 100.211(b)(3), a pilot who does not complete the instrument practical test and remaining ACS tasks of the type rating practical test within 2 calendar months from issuance of the initial powered-lift type rating will no longer be qualified to fly a powered-lift. In order to restore their powered-lift privileges, the pilot must have 3 hours of training within the previous 2 calendar months and complete the requisite practical tests.

5.8.2 If a pilot does not complete the instrument practical test in the powered-lift within 2 calendar months and that pilot is qualified in another category aircraft, they may continue to exercise their other privileges while their powered-lift privileges are invalid. For example, if a pilot holds an airplane category rating and is adding a powered-lift category rating and initial type rating but does not complete the practical test to remove the “VFR only” limitation within 2 calendar months, they can continue to exercise their airplane privileges.

5.9 Obtaining Additional Powered-Lift Type Ratings With “VFR Only” Limitation. A person holding a Private Pilot Certificate may apply for an additional powered-lift type rating with a “VFR only” limitation. Pursuant to GACAR § 100.211(c), an applicant who holds a Private Pilot Certificate and is seeking an additional powered-lift type rating for a powered-lift that is not a large aircraft or turbojet-powered need not comply with the requirements of GACAR § 61.83(d)(1). Rather, the applicant may apply without holding or concurrently obtaining an instrument-powered-lift rating, but the type rating will be limited to “VFR only.” As with obtaining an initial powered-lift type rating with a “VFR only” limitation, the applicant does not have to take the type rating practical test in actual or simulated instrument conditions and may remove the limitation for that powered-lift in the same manner specified to remove a “VFR only” limitation on an initial powered-lift type rating, as set forth in GACAR § 100.211(b)(4) (and discussed in paragraph 5.7 of this AC).

CHAPTER 6. ALTERNATE AERONAUTICAL EXPERIENCE AND LOGGING REQUIREMENTS—GENERAL PROVISIONS

6.1 Alternate Means to Obtain Ratings.

6.1.1 Pilots seeking to receive a Commercial Pilot Certificate with a powered-lift category rating in accordance with Appendix D, paragraph (e) to GACAR Part 61 and an instrument-powered-lift rating in accordance with Appendix E, paragraph (c) to GACAR Part 61 may encounter certain challenges in gaining the aeronautical experience necessary to receive these ratings. Therefore, the GACA has established alternate means for certain persons seeking to obtain ratings in a powered-lift. A person may choose to comply with the alternate aeronautical experience and logging requirements or the standard GACAR Part 61 requirements. However, if no alternate aeronautical experience or logging requirement is provided in GACAR Part 100, then the person must meet the applicable GACAR Part 61 requirements, pursuant to GACAR § 100.215(b).

6.1.2 GACAR Part 100 provides a pathway for new aircraft entering the civilian market and the initial pilots who fly these aircraft. The alternate pathway is broken down into three groups with specific alternate aeronautical experience requirements set forth to qualify the person to take the practical test. The three groups include:

1. *Instructor pilots and test pilots at a manufacturer.* Test pilots and instructor pilots employed by a specific manufacturer for aircraft undergoing certification are authorized to log aeronautical experience in these aircraft through a Letter of Authorization (LOA) issued by the President. Additionally, CAA test pilots or aviation safety inspectors (ASI) will also be able to log aeronautical experience in these aircraft when conducting duties for the purpose of establishing new type designs in the case of a CAA test pilot, or for the purpose of establishing a pilot type rating in the case of an ASI.
2. *Initial cadre of instructors at a certificate holder (CH) (check pilots, chief instructors, assistant chief instructors, or Training Center Evaluators (TCE)).* These individuals will need to satisfactorily complete the manufacturer's training curriculum in a type-certificated aircraft at the manufacturer to qualify for the practical test to add a powered-lift category rating.
3. *Pilots receiving training under an approved training program.* CHs will need to apply for and receive approval of a training curriculum so that the initial cadre of instructors who train with the manufacturer can return to the CH and train pilots on the use of the powered-lift.

Note: The GACA uses the term “initial cadre” throughout this AC. In some instances, initial cadre refers to a sufficient number of instructors and evaluators to train and qualify pilots for

powered-lift ratings under an approved training program under GACAR Part 135, 141, or 142. In other instances, the term refers to a sufficient number of pilots who are rated in powered-lift.

6.1.3 In accordance with GACAR § 100.215(a), the alternate aeronautical experience requirements (except for cross-country) for a Commercial Pilot Certificate and instrument-powered-lift rating are only applicable to individuals who hold at least a Commercial Pilot Certificate with:

1. An airplane category rating with a single-engine or multiengine class rating and an instrument-airplane rating; or
2. A rotorcraft category rating with a helicopter class rating and an instrument-helicopter rating.

6.1.4 Pilots seeking a Commercial Pilot Certificate with a powered-lift category rating in accordance with GACAR § 100.216(a) may log 35 hours of pilot-in-command (PIC) flight time in a powered-lift in lieu of the 50 hours of PIC flight time required by Appendix D, paragraph (e)(1)(i) to GACAR Part 61.

6.1.5 Additionally, under GACAR § 100.216(b), applicants utilizing a Level C full flight simulator (FFS) or higher may credit 15 hours toward the 35-hour PIC flight time so long as:

1. The aeronautical experience was obtained in a Level C FFS or higher representing a powered-lift; and
2. The FFS sessions are conducted in accordance with:
 - a. A manufacturer's proposed training curriculum for test pilots, instructor pilots, CAA test pilots, or ASIs under GACAR §§ 100.217 or 100.219;
 - b. The manufacturer's training curriculum for the initial cadre of instructors under GACAR § 100.221; or
 - c. An approved training program under GACAR Part 135, 141, or 142.

CHAPTER 7. ALTERNATE AERONAUTICAL AND LOGGING EXPERIENCE REQUIREMENTS FOR A COMMERCIAL PILOT CERTIFICATE WITH A POWERED-LIFT CATEGORY RATING

7.1 Alternate Provisions for Test Pilots, CAA Test Pilots, Aviation Safety Inspectors (ASI), and Instructor Pilots.

7.1.1 *Initial Experience During Aircraft Certification.* GACAR §§ 100.217 and 100.219 provide alternate experience requirements for test pilots and instructor pilots of the manufacturer, as well as CAA test pilots and ASIs. To utilize these alternate experience requirements, the flights must be conducted in an experimental powered-lift at the manufacturer, and the applicant (i.e., the test pilot, CAA test pilot, ASI, or instructor pilot) must be authorized by the President to act as pilot in command (PIC) of the experimental powered-lift. The specific alternate requirements are:

- PIC flight time;
- Training on the areas of operation listed in Appendix C, Section V, paragraph (d)(5) of GACAR Part 141 in a powered-lift at the manufacturer; and
- Solo flight time or flight time performing the duties of PIC.

7.1.2 *PIC Flight Time.* Pursuant to GACAR §§ 100.217(c) and 100.219(c), a test pilot, a CAA test pilot, an ASI, or an instructor pilot may log PIC flight time for the purpose of satisfying the aeronautical experience requirements in Appendix D, paragraph (e)(2)(ii) to GACAR Part 61 and GACAR § 100.216(a) when the following conditions are met:

1. The instructor pilot, CAA test pilot, ASI, or test pilot is the sole manipulator of the controls of an experimental powered-lift for which the pilot is not rated;
2. The instructor pilot, CAA test pilot, ASI, or test pilot is acting as PIC of the experimental powered-lift in accordance with a Letter of Authorization (LOA) issued by the President;
3. In the case of a test pilot, a CAA test pilot, or an ASI, the flight is conducted for the purposes of:
 - (a) research and development (R&D), or
 - (b) showing compliance with the regulations in accordance with the experimental certificate issued to the powered-lift; and
4. In the case of an instructor pilot, the flight is conducted for the purposes of crew training in accordance with the experimental certificate issued to the powered-lift.

7.1.3 *Training on the Areas of Operation Listed in Appendix C, Section V, paragraph (d)(5) of GACAR Part 141 in a Powered-Lift at the Manufacturer.*

7.1.3.1 An applicant for a Commercial Pilot Certificate with a powered-lift category rating who has not received the 20 hours of flight training on the areas of operation listed in Appendix C, Section V, paragraph (d)(5) of GACAR Part 141 from an authorized instructor may be considered eligible under the following two circumstances, as specified in GACAR § 100.217. In accordance with GACAR §§ 100.217(b)(1) and 100.219(b)(1), the applicant must have satisfactorily completed the manufacturer's proposed training curriculum. In the case of a test pilot, a CAA test pilot, or an ASI applicant, the training curriculum must have been provided by a pilot who serves as an instructor pilot in the experimental powered-lift. In the case of an instructor pilot applicant, the person must have provided the manufacturer's proposed training curriculum in the experimental powered-lift. In either case, the applicant must receive a logbook or training record endorsement. In the case of a test pilot, a CAA test pilot, or an ASI applicant, this endorsement must be from an instructor pilot documenting completion of the manufacturer's training curriculum. In the case of an instructor pilot, the endorsement must be from a management official within the manufacturer's organization certifying the instructor pilot provided the training in accordance with the manufacturer's proposed training curriculum. An example of the training record endorsement can be found in Appendix A, Example Endorsements.

7.1.3.2 GACAR § 100.217(b)(2) and § 100.219(b)(2) permit an applicant for a Commercial Pilot Certificate with powered-lift category rating to be eligible for a certificate if they receive 3 hours of flight training in the experimental powered-lift in preparation for the practical test within the preceding 2 calendar months from the month of the test, if the training is received from an instructor pilot employed by the manufacturer of the experimental powered-lift.

7.1.3.3 Pursuant to GACAR §§ 100.217(b)(4) and 100.219(b)(4), test pilots, CAA test pilots, ASIs, or instructor pilots who do not meet the cross-country aeronautical experience requirements specified in Appendix D, paragraph (e)(3)(ii) and (iii) to GACAR Part 61 may be considered eligible for a Commercial Pilot Certificate if they complete the cross-country requirements set forth by GACAR § 100.233(a). These requirements are detailed in Chapter 3, Alternate Cross-Country Requirements. The applicant must obtain a logbook or training record endorsement, as specified in GACAR § 100.213, from the instructor pilot or management official, as appropriate, certifying that the person completed the cross-country flight.

7.1.4 Solo Flight Time or Flight Time Performing the Duties of PIC.

7.1.4.1 Each applicant is required to meet the requirements of Appendix D, paragraph (e)(4) to GACAR Part 61. In order to do this in a powered-lift that requires two pilots, GACAR §§ 100.217(b)(3) and 100.219(b)(3) enable alternative requirements. Specifically, a test pilot, a CAA test pilot, an ASI, or an instructor pilot applicant may log at least 10 hours of solo flight time under an endorsement from an instructor pilot or by performing the duties of PIC in an experimental powered-lift with either:

- (a) a test pilot for the manufacturer, or

(b) an instructor pilot for the manufacturer. In either case, the additional test pilot or the instructor pilot must be authorized to act as the PIC of the experimental powered-lift.

7.1.4.2 When an applicant does not have the cross-country aeronautical experience specified in Appendix D, paragraph (e)(4)(i) to GACAR Part 61, the applicant will still be considered eligible for a Commercial Pilot Certificate with powered-lift category rating if the applicant has logged at least two cross-country flights with landings at a minimum of three points, with one segment consisting of a straight-line distance of at least 50 nautical miles (NM) from the original point of departure. GACAR § 100.233(b) details that the second cross-country flight, besides the original point of departure, must be performed to different points than the first cross-country flight.

7.1.4.3 In order to demonstrate compliance with the alternate requirements provided by GACAR Part 100, the applicant must log this cross-country experience in their logbook as they would for any other certificate or rating in accordance with GACAR § 61.13(a).

Table 7-1. Alternate Provisions for a Commercial Pilot Certificate With a Powered-Lift Category Rating: Test Pilots and Instructor Pilots

CURRENT FLIGHT TIME REQUIREMENTS (Appendix D, paragraph (e) to GACAR Part 61)	ALTERNATE PROVISIONS FOR TEST PILOTS, CAA TEST PILOTS, OR ASIs*	ALTERNATE PROVISIONS FOR INSTRUCTOR PILOTS*
*To use the alternate provisions below, the flights must be conducted in an experimental powered-lift at the manufacturer, and the test pilots and instructor pilots must be authorized by the President to act as PIC of the experimental powered-lift. Refer to GACAR §§ 100.217(a) and 100.219(a).		
Powered-Lift Aircraft Time		
100 hours in powered aircraft, of which 50 hours must be in a powered-lift.	No alternative.	No alternative.
PIC Flight Time		
100 hours of PIC flight time, which includes at least: (1) 50 hours in a powered-lift, and (2) 50 hours in cross-country flight of which 10 must be in a powered-lift.	100 hours of PIC flight time, which includes at least: (1) 35 hours in a powered-lift of which 15 may be in a Level C or higher full flight simulator (FFS), and (2) 50 hours in cross-country flight of which 10 must be in a powered-lift. A test pilot, a CAA test pilot, or an ASI who is not rated in the powered-lift may log PIC flight time in the experimental powered-lift if: (1) The pilot is the sole manipulator of the controls, (2) The pilot is acting as PIC, and	100 hours of PIC flight time, which includes at least: (1) 35 hours in a powered-lift of which 15 may be in a Level C or higher FFS, and (2) 50 hours in cross-country flight of which 10 must be in a powered-lift. An instructor pilot who is not rated in the powered-lift may log PIC flight time in the experimental powered-lift if: (1) The instructor pilot is acting as PIC, and (2) The flight is conducted for the purpose of crew training. Refer to GACAR §§ 100.216 and 100.219(c).

CURRENT FLIGHT TIME REQUIREMENTS (Appendix D, paragraph (e) to GACAR Part 61)	ALTERNATE PROVISIONS FOR TEST PILOTS, CAA TEST PILOTS, OR ASIs*	ALTERNATE PROVISIONS FOR INSTRUCTOR PILOTS*
	(3) The flight is conducted for the purpose of R&D or showing compliance. Refer to GACAR §§ 100.216 and 100.217(c).	
Training on the Areas of Operation Listed in Appendix C, Section V, paragraph (d)(5) of GACAR Part 141		
20 hours of training on the areas of operation listed in Appendix C, Section V, paragraph (d)(5) of GACAR Part 141 that includes at least:	A test pilot, a CAA test pilot, or an ASI may receive this training from an instructor pilot (rather than an authorized instructor) if: (1) Training is conducted in accordance with the manufacturer's training curriculum, and (2) The test pilot, CAA test pilot, or ASI receive endorsement from the instructor pilot certifying satisfactory completion of the curriculum. Refer to GACAR § 100.217(b)(1).	In lieu of receiving this training, an instructor pilot may meet this requirement by providing the manufacturer's training curriculum to a test pilot, a CAA test pilot, or an ASI, which includes 20 hours of training on the areas of operation listed in Appendix C, Section V, paragraph (d)(5) of GACAR Part 141. The instructor pilot must receive endorsement from a management official certifying that the instructor pilot provided the training. Refer to GACAR § 100.219(b)(1).
(1) 10 hours of instrument training using a view-limiting device, including attitude instrument flying, partial panel skills, recovery from unusual flight attitudes, and intercepting and tracking navigational systems. 5 hours of the 10 hours required on instrument training must be in a powered-lift.	A test pilot, a CAA test pilot, or an ASI may receive this instrument training from an instructor pilot in an experimental powered-lift in accordance with the manufacturer's proposed training curriculum. Refer to § 100.217(b)(1).	An instructor pilot may satisfy this requirement by providing instrument training to a test pilot, a CAA test pilot, or an ASI in accordance with the manufacturer's proposed training curriculum. The instructor pilot must receive an endorsement from a management official certifying that the instructor pilot provided the training. Refer to § 100.219(b)(1).
(2) The following cross-country flights: • One 2-hour cross-country flight in a powered-lift in daytime conditions that consists of a total straight-line distance of more than 100 NM from the original point of departure; and • One 2-hour cross-country flight in a powered-lift in nighttime conditions that consists of a total	A test pilot, a CAA test pilot, or an ASI may complete the following cross-country flights: • One 2-hour cross-country flight in a powered-lift in daytime conditions that consists of a total straight-line distance of more than 50 NM from the original point of departure; • One 2-hour cross-country flight in a powered-lift in nighttime conditions that consists of a total	An instructor pilot may complete the same alternate cross-country flights as a test pilot, a CAA test pilot, or an ASI. Refer to § 100.233(a).

CURRENT FLIGHT TIME REQUIREMENTS (Appendix D, paragraph (e) to GACAR Part 61)	ALTERNATE PROVISIONS FOR TEST PILOTS, CAA TEST PILOTS, OR ASIs*	ALTERNATE PROVISIONS FOR INSTRUCTOR PILOTS*
straight-line distance of more than 100 NM from the original point of departure.	straight-line distance of more than 50 NM from the original point of departure; and • An additional cross-country flight with landings at a minimum of three points, with one segment consisting of a straight-line distance of at least 50 NM from the original point of departure. The additional cross-country flight must include landings at different points than the first two cross-country flights. Refer to § 100.233(a).	
(3) 3 hours in a powered-lift with an authorized instructor in preparation for the practical test within the preceding 2 calendar months from the month of the test.	A test pilot, a CAA test pilot, or an ASI may accomplish the practical test preparation with an instructor pilot (rather than an authorized instructor). Refer to § 100.217(b)(2).	An instructor pilot may accomplish the practical test preparation with another instructor pilot (rather than an authorized instructor). Refer to § 100.219(b)(2).
Solo Flight Time or Flight Time Performing the Duties of PIC		
10 hours of solo flight time in a powered-lift or 10 hours of flight time performing the duties of PIC in a powered-lift with an authorized instructor on board (either of which may be credited toward the flight time requirement under Appendix D, paragraph (e)(2) to GACAR Part 61 on the areas of operation listed in Appendix C, Section V, paragraph (d)(5) of GACAR Part 141 that includes:	A test pilot, a CAA test pilot, or an ASI may complete 10 hours of solo flight time under an endorsement from an instructor pilot or 10 hours of flight time performing the duties of PIC in a powered-lift with either a test pilot or an instructor pilot on board. Refer to § 100.217(b)(3).	The alternate provision for test pilots, CAA test pilots, or ASIs also applies to instructor pilots. Refer to § 100.219(b)(3).
(1) One cross-country flight of not less than 300 NM total distance with landings at a minimum of three points, one of which is a straight-line distance of at least 250 NM from the original departure point.	A test pilot, a CAA test pilot, or an ASI may complete two cross-country flights with landings at a minimum of three points, with one segment consisting of a straight-line distance of at least 50 NM from the original point of departure. The second cross-country flight must include landings at different points than the first cross-country flight. Refer to § 100.233(b).	An instructor pilot may complete the same alternate cross-country flights as a test pilot, a CAA test pilot, or an ASI. Refer to § 100.233(b).

CURRENT FLIGHT TIME REQUIREMENTS (Appendix D, paragraph (e) to GACAR Part 61)	ALTERNATE PROVISIONS FOR TEST PILOTS, CAA TEST PILOTS, OR ASIs*	ALTERNATE PROVISIONS FOR INSTRUCTOR PILOTS*
(2) 5 hours in night visual flight rules (VFR) conditions with 10 takeoffs and 10 landings (with each landing involving a flight in the traffic pattern) at an aerodrome with an operating control tower.	No alternative.	No alternative.

7.2 Alternate Provisions for Initial Cadre of Instructors.

7.2.1 Facilitation of Training. Once the manufacturer has completed type certification of their powered-lift, GACAR Part 100 facilitates manufacturers providing training for an initial cadre of instructors employed at specific certificate holders (CH). The manufacturer's training curriculum will cover all the required areas of operation for certification, and the manufacturer's instructor pilots may deliver this training. This approach of facilitating the use of instructor pilots at the manufacturer to deliver initial certification training to the first powered-lift applicants enables those most familiar with the aircraft (i.e., the manufacturers) to provide safe and effective training and provides the manufacturer the means to ensure that there are a sufficient number of qualified pilots to operate their aircraft and train other pilots to operate their aircraft.

7.2.2 Authorization. In accordance with GACAR § 100.221(a), the initial cadre instructor applicant must be authorized by the President to serve as an initial check pilot, a chief instructor, an assistant chief instructor, or a Training Center Evaluator (TCE) for the purpose of initiating training in a powered-lift under an approved training program under GACAR Part 135, 141, or 142, as appropriate. The CH should submit a request to GACA. Upon authorization, these applicants will receive a temporary LOA from the President to receive this training. As set forth by GACAR § 100.221 and subsequently explained in the paragraphs below, the applicant must satisfactorily complete the training curriculum, and the training must be provided by the manufacturer in a type-certificated powered-lift with a pilot who serves as an instructor pilot for that manufacturer.

7.2.2.1 Initial Experience for Initial Cadre Instructors. GACAR § 100.221 sets forth the alternate experience requirements for the initial cadre of instructors. These specific alternate requirements are:

- PIC flight time,
- Training on the areas of operation listed in Appendix C, Section V, paragraph (d)(5) of GACAR Part 141 in a powered-lift at the manufacturer, and
- Solo flight time or flight time performing the duties of PIC.

7.2.2.2 *PIC Flight Time.* GACAR § 100.221(c) authorizes an initial cadre instructor applicant who is not rated in the powered-lift to log up to 25 hours of the required 35 hours of PIC flight time in a powered-lift for flights conducted in accordance with a manufacturer's training curriculum if the applicant is:

- The sole manipulator of the controls,
- Manipulating the controls of the powered-lift with an instructor pilot on board, and
- Performing the duties of PIC.

Note: The last 10 hours of the required 35 hours must be either solo or performing the duties of PIC with an authorized instructor or instructor pilot on board.

7.2.2.3 *Training on the Areas of Operation Listed in Appendix C, Section V, paragraph (d)(5) of GACAR Part 141 in a Powered-Lift at the Manufacturer.*

7.2.2.3.1 GACAR § 100.221(b)(1) authorizes an initial cadre instructor applicant to receive training from an instructor pilot (rather than an authorized instructor) in a type-certificated powered-lift at a manufacturer in accordance with the manufacturer's training curriculum, if the curriculum includes 20 hours of training on the areas of operation listed in Appendix C, Section V, paragraph (d)(5) of GACAR Part 141.

7.2.2.3.2 GACAR § 100.221(b)(2) permits an initial cadre instructor applicant to receive 3 hours of flight training in the type-certificated powered-lift in preparation for the practical test within the preceding 2 calendar months from the month of the test, if it is received from an instructor pilot employed by the manufacturer.

7.2.2.3.3 Pursuant to GACAR § 100.221(b)(4), an initial cadre instructor applicant who does not meet the cross-country aeronautical experience requirements specified in Appendix D, paragraph (e)(3)(ii) and (iii) to GACAR Part 61 will still be considered eligible for a Commercial Pilot Certificate if they complete the cross-country requirements listed in GACAR § 100.233(a). These requirements are detailed in Chapter 3 of this AC.

7.2.2.3.4 The initial cadre instructor applicant must obtain a logbook or training record endorsement, as specified in GACAR § 100.213, from the instructor pilot certifying that the person completed the cross-country flight.

7.2.2.4 *Solo Flight Time or Flight Time Performing the Duties of PIC.*

7.2.2.4.1 Each initial cadre instructor applicant must meet the requirements of Appendix D, GACAR Part 61. These requirements include completing 10 hours of solo flight time in a powered-lift. Because solo flight time is not possible in a powered-lift that requires two pilots, GACAR § 100.221(b)(3) established an alternative requirement. Specifically, an applicant in the initial cadre group may log at least 10 hours of solo flight time in a powered-lift under an

endorsement from an instructor pilot or by performing the duties of the PIC with an instructor pilot on board.

7.2.2.4.2 If an initial cadre instructor applicant does not have the cross-country aeronautical experience specified in Appendix D, paragraph (e)(4)(i) to GACAR Part 61, the applicant may be considered eligible for a Commercial Pilot Certificate with powered-lift category rating if the applicant has logged at least two cross-country flights with landings at a minimum of three points, with one segment consisting of a straight-line distance of at least 50 NM from the original point of departure. GACAR § 100.233(b) details that the second cross-country flight, besides the original point of departure, must be performed to different points than the first cross-country flight.

7.2.2.4.3 To demonstrate compliance with the alternate requirements provided by GACAR Part 100, the initial cadre instructor applicant must log this cross-country experience in their logbook as they would for any other certificate.

7.2.2.4.4 Once the initial cadre instructor applicant completes the manufacturer's training curriculum, they will receive a logbook or training record endorsement from the instructor pilot certifying that they satisfactorily completed the training in accordance with GACAR § 100.221(b)(1)(ii).

Table 7-2. Alternate Provisions for a Commercial Pilot Certificate with a Powered-Lift Category Rating: Initial Cadre of Check Pilots, Chief Instructors, Assistant Chief Instructors, and Training Center Evaluators (Initial Cadre Instructors)

CURRENT FLIGHT TIME REQUIREMENTS (Appendix D, to GACAR Part 61)	ALTERNATE PROVISIONS FOR INITIAL CADRE OF CHECK PILOTS, CHIEF INSTRUCTORS, ASSISTANT CHIEF INSTRUCTORS, AND TCEs* (initial cadre instructors, herein referred to as applicants)
*To use the alternate provisions below, the flights must be conducted by an instructor pilot in a type certificated powered-lift at the manufacturer, and the applicant must be authorized by the President to serve as an initial check pilot, a chief instructor, an assistant chief instructor, or a TCE for the purpose of initiating training in a powered-lift under an approved training program under GACAR Part 135, 141, or 142, as appropriate. Refer to GACAR § 100.221(a).	
Powered-Lift Aircraft Time	
100 hours in powered aircraft, of which 50 hours must be in a powered-lift.	No alternative.
PIC Flight Time	
100 hours of PIC flight time, which includes at least: (1) 50 hours in a powered-lift, and (2) 50 hours in cross-country flight of which 10 must be in a powered-lift.	100 hours of PIC flight time, which includes at least: (1) 35 hours in a powered-lift, 15 of which may be in a Level C or higher FFS, and (2) 50 hours in cross-country flight of which 10 must be in a powered-lift. An applicant who is not rated in the powered-lift may log up to 25 hours of PIC flight time in a powered-lift

CURRENT FLIGHT TIME REQUIREMENTS (Appendix D, to GACAR Part 61)	ALTERNATE PROVISIONS FOR INITIAL CADRE OF CHECK PILOTS, CHIEF INSTRUCTORS, ASSISTANT CHIEF INSTRUCTORS, AND TCEs* (initial cadre instructors, herein referred to as applicants)
	for flights that are conducted in accordance with a manufacturer's training curriculum if the applicant is: (1) The sole manipulator of the controls, (2) Manipulating the controls of the powered-lift with an instructor pilot on board, and (3) Performing the duties of PIC. Refer to §§ 100.216 and 100.221(c).
Training on the Areas of Operation Listed in Appendix C, Section V, paragraph (d)(5) of GACAR Part 141	
20 hours of training on the areas of operation listed in Appendix C, Section V, paragraph (d)(5) of GACAR Part 141 that includes at least:	An applicant may receive this training from an instructor pilot (rather than an authorized instructor) if: (1) Training is conducted in accordance with the manufacturer's training curriculum, and (2) The applicant receives an endorsement from an instructor pilot certifying satisfactory completion of the curriculum. Refer to § 100.221(b)(1).
(1) 10 hours of instrument training using a view-limiting device, including attitude instrument flying, partial panel skills, recovery from unusual flight attitudes, and intercepting and tracking navigational systems. 5 hours of the 10 hours required on instrument training must be in a powered-lift.	An applicant may receive this instrument training from an instructor pilot in a type-certificated powered-lift in accordance with the manufacturer's training curriculum. Refer to § 100.221(b)(1).
(2) The following cross-country flights: • One 2-hour cross-country flight in a powered-lift in daytime conditions that consists of a total straight-line distance of more than 100 NM from the original point of departure; and • One 2-hour cross-country flight in a powered-lift in nighttime conditions that consists of a total straight-line distance of more than 100 NM from the original point of departure.	An applicant may complete the following cross-country flights: • One 2-hour cross-country flight in a powered-lift in daytime conditions that consists of a total straight-line distance of more than 50 NM from the original point of departure; • One 2-hour cross-country flight in a powered-lift in nighttime conditions that consists of a total straight-line distance of more than 50 NM from the original point of departure; and • An additional cross-country flight with landings at a minimum of three points, with one segment consisting of a straight-line distance of at least 50 NM from the original point of departure. The additional cross-country flight must include landings at different points than the first two cross-country flights. Refer to § 100.233(a).
(3) 3 hours in a powered-lift with an authorized instructor in preparation for the practical test within the preceding 2 calendar months from the month of the test.	An applicant may accomplish the practical test preparation from an instructor pilot (rather than an authorized instructor). Refer to § 100.221(b)(2).
Solo Flight Time or Flight Time Performing the Duties of PIC	

CURRENT FLIGHT TIME REQUIREMENTS (Appendix D, to GACAR Part 61)	ALTERNATE PROVISIONS FOR INITIAL CADRE OF CHECK PILOTS, CHIEF INSTRUCTORS, ASSISTANT CHIEF INSTRUCTORS, AND TCEs* (initial cadre instructors, herein referred to as applicants)
10 hours of solo flight time in a powered-lift or 10 hours of flight time performing the duties of PIC in a powered-lift with an authorized instructor on board (either of which may be credited toward the flight time requirement under Appendix D, paragraph (e)(2) to GACAR Part 61 on the areas of operation listed in Appendix C, Section V, paragraph (d)(5) of GACAR Part 141 that includes:	An applicant may complete 10 hours of solo flight time under an endorsement from an instructor pilot or 10 hours of flight time performing the duties of PIC in a powered-lift with an instructor pilot on board (rather than an authorized instructor). Refer to § 100.221(b)(3).
(1) One cross-country flight of not less than 300 NM total distance with landings at a minimum of three points, one of which is a straight-line distance of at least 250 NM from the original departure point.	An applicant may complete two cross-country flights with landings at a minimum of three points, with one segment consisting of a straight-line distance of at least 50 NM from the original point of departure. The second cross-country flight must include landings at different points than the first cross-country flight. Refer to § 100.233(b).
(2) 5 hours in night VFR conditions with 10 takeoffs and 10 landings (with each landing involving a flight in the traffic pattern) at an aerodrome with an operating control tower.	No alternative.

7.3 Alternate Provisions for Pilots Trained in an Approved Training Program at the CH. Newly certificated initial cadre flight instructors for an operator, a pilot school, or a training center (i.e., GACAR Part 135 check pilots, Part 141 chief instructors or assistant chief instructors, or Part 142 TCEs who have been trained and certified at the manufacturer) will be proficient in the training curriculum under their respective part because of their involvement in developing the proposed training program. This involvement will enable them to be in the best position to provide training to pilots enrolled in their respective CH's training program under GACAR Part 135, 141, or 142. The initial cadre of flight instructors will become fully qualified to serve in their designated function and receive a permanent LOA from the President to subsequently conduct the training. Because these pilots will now be authorized instructors (i.e., meet the requirements for instructors under their respective part), the pilots who will receive training under an approved training program do not require the same breadth of relief that instructor pilots, test pilots, and the initial cadre instructors need. Rather, the only relief necessary for pilots enrolled in an approved training program is that of logging, flight simulation training device (FSTD) credit, and the cross-country relief in GACAR § 100.233 extended to all applicants, which would require the trainer to hold a flight instructor certificate. See Chapter 12, Requirements to Serve as a Powered-Lift Pilot Under Part 135, for additional information.

7.3.1 PIC Flight Time.

7.3.1.1 GACAR § 100.223(c) authorizes an applicant who is not rated in the powered-lift to log up to 25 hours of the required 35 hours of PIC flight time in a powered-lift for flights that are conducted in accordance with an approved training program under GACAR Part 135, 141, or 142 if the applicant is:

- The sole manipulator of the controls,
- Manipulating the controls of the powered-lift with an authorized instructor on board, and
- Performing the duties of PIC.

Note: The last 10 hours of the required 35 hours must be either solo or performing the duties of PIC with an authorized instructor or instructor pilot on board.

7.3.1.2 GACAR § 100.216 allows an applicant to credit up to 15 hours of flight time obtained in a Level C or higher FFS toward the requirement to obtain 35 hours of PIC flight time in a powered-lift if:

- The FFS represents the powered-lift category, and
- The applicant was performing the duties of PIC.

7.3.2 *Training on the Areas of Operation Listed in Appendix C, Section V, paragraph (d)(5) of GACAR Part 141 in a Powered-Lift.* Applicants who do not meet the cross-country aeronautical experience requirements specified in Appendix D, paragraph (e)(3)(ii) and (iii) to GACAR Part 61 may be considered eligible for a Commercial Pilot Certificate if they complete the cross-country requirements as specified in GACAR § 100.233(a). These requirements are detailed in Chapter 3 of this AC.

7.3.3 *Solo Flight Time or Flight Time Performing the Duties of PIC.* If an applicant does not have the cross-country aeronautical experience specified in Appendix D, paragraph (e)(4)(i) to GACAR Part 61, the applicant may be considered eligible for a Commercial Pilot Certificate with powered-lift category rating if the applicant has logged at least two cross-country flights with landings at a minimum of three points, with one segment consisting of a straight-line distance of at least 50 NM from the original point of departure. GACAR § 100.233(b) details that the second cross-country flight, besides the original point of departure, must be performed to different points than the first cross-country flight.

Table 7-3. Alternate Provisions for a Commercial Pilot Certificate with a Powered-Lift Category Rating: Persons Receiving Training Under a Part 135, 141, or 142 Approved Training Program

CURRENT FLIGHT TIME REQUIREMENTS (Appendix D, to GACAR Part 61)	ALTERNATE PROVISIONS FOR PERSONS RECEIVING TRAINING UNDER A GACAR PART 135, 141, OR 142 APPROVED TRAINING PROGRAM* (herein referred to as applicants)
*To use the alternate provisions below, the applicant must receive training at an approved training program under GACAR Part 135, 141, or 142 for the purpose of obtaining a powered-lift category rating.	
Powered-Lift Aircraft Time	
100 hours in powered aircraft, of which 50 hours must be in a powered-lift.	No alternative.
PIC Flight Time	
100 hours of PIC flight time, which includes at least: (1) 50 hours in a powered-lift, and (2) 50 hours in cross-country flight of which 10 must be in a powered-lift.	100 hours of PIC flight time, which includes at least: (1) 35 hours in a powered-lift, 15 of which may be in a Level C or higher FFS, and (2) 50 hours in cross-country flight of which 10 must be in a powered-lift. An applicant who is not rated in the powered-lift may log up to 25 hours of PIC flight time in a powered-lift for flights that are conducted in accordance with an approved training program under GACAR Part 135, 141, or 142 if the applicant is: (1) The sole manipulator of the controls, (2) Manipulating the controls of the powered-lift with an authorized instructor on board, and (3) Performing the duties of PIC. Refer to §§ 100.216 and 100.223(c).
Training on the Areas of Operation Listed in Appendix C, Section V, paragraph (d)(5) of GACAR Part 141	
20 hours of training on the areas of operation listed in Appendix C, Section V, paragraph (d)(5) of GACAR Part 141 that includes at least:	No alternative.
(1) 10 hours of instrument training using a view-limiting device, including attitude instrument flying, partial panel skills, recovery from unusual flight attitudes, and intercepting and tracking navigational systems. 5 hours of the 10 hours required on instrument training must be in a powered-lift.	No alternative.
(2) The following cross-country flights: • One 2-hour cross-country flight in a powered-lift in daytime conditions that consists of a total straight-line distance of more than 100 NM from the original point of departure; and • One 2-hour cross-country flight in a powered-lift in nighttime conditions that consists of a total straight-line distance of more than 100 NM from the original point of departure.	An applicant may complete the following cross-country flights: • One 2-hour cross-country flight in a powered-lift in daytime conditions that consists of a total straight-line distance of more than 50 NM from the original point of departure; • One 2-hour cross-country flight in a powered-lift in nighttime conditions that consists of a total straight-line distance of more than 50 NM from the original point of departure; and • An additional cross-country flight with landings at a minimum of three points, with one segment consisting

	of a straight-line distance of at least 50 NM from the original point of departure. The additional cross-country flight must include landings at different points than the first two cross-country flights. Refer to § 100.233(a).
(3) 3 hours in a powered-lift with an authorized instructor in preparation for the practical test within the preceding 2 calendar months from the month of the test.	No alternative.
Solo Flight Time or Flight Time Performing the Duties of PIC	
10 hours of solo flight time in a powered-lift or 10 hours of flight time performing the duties of PIC in a powered-lift with an authorized instructor on board (either of which may be credited toward the flight time requirement under Appendix D, paragraph (e)(2) to GACAR Part 61 on the areas of operation listed in Appendix C, Section V, paragraph (d)(5) of GACAR Part 141 that includes:	No alternative.
(1) One cross-country flight of not less than 300 NM total distance with landings at a minimum of three points, one of which is a straight-line distance of at least 250 NM from the original departure point.	An applicant may complete two cross-country flights with landings at a minimum of three points, with one segment consisting of a straight-line distance of at least 50 NM from the original point of departure. The second cross-country flight must include landings at different points than the first cross-country flight. Refer to § 100.233(b).
(2) 5 hours in night VFR conditions with 10 takeoffs and 10 landings (with each landing involving a flight in the traffic pattern) at an aerodrome with an operating control tower.	No alternative.

CHAPTER 8. ALTERNATE AERONAUTICAL EXPERIENCE AND LOGGING REQUIREMENTS FOR AN INSTRUMENT-POWERED-LIFT RATING

8.1 Alternate Provisions for Test Pilots, CAA test pilots, Aviation Safety Inspectors (ASI), and Instructor Pilots.

8.1.1 *Initial Experience During Aircraft Certification.* GACAR §§ 100.225 and 100.227 provide certain alternate experience requirements for an instrument-powered-lift rating for test pilots, CAA test pilots, ASIs, and instructor pilots. To utilize these alternate experience requirements, the flights must be conducted at an experimental powered-lift at the manufacturer, and the applicant (i.e., the test pilot, CAA test pilot, ASI, or instructor pilot) must be authorized by the President to act as pilot in command (PIC) of the experimental powered-lift. The alternate experience requirements include cross-country time as PIC, and instrument time on the areas of operation listed in Section IV(d) of Appendix B to GACAR Part 141.

8.1.2 *Alternate Logging Requirements.* Pursuant to GACAR §§ 100.225(c) and 100.227(c), a test pilot, a CAA test pilot, an ASI, or an instructor pilot may log PIC flight time for the purpose of satisfying the aeronautical experience requirements in Appendix E, paragraph (c)(1) to GACAR Part 61 when the following conditions are met:

1. The instructor pilot, CAA test pilot, ASI, or test pilot is the sole manipulator of the controls of an experimental powered-lift for which the pilot is not rated;
2. The instructor pilot, CAA test pilot, ASI, or test pilot is acting as PIC of the experimental powered-lift in accordance with a Letter of Authorization (LOA) issued by the President;
3. In the case of a test pilot, a CAA test pilot, or an ASI, the flight is conducted for the purposes of:
 - (a) research and development (R&D), or
 - (b) showing compliance with the regulations in accordance with the experimental certificate issued to the powered-lift; and
4. In the case of an instructor pilot, the flight is conducted for the purposes of crew training in accordance with the experimental certificate issued for the powered-lift.

8.1.3 *Instrument Training on the Areas of Operation Listed in Section IV(d) of Appendix B to GACAR Part 141.*

8.1.3.1 An applicant for an instrument-powered-lift rating who has not received the 15 hours of instrument training on the areas of operation listed in Section IV(d) of Appendix B to GACAR Part 141 from an authorized instructor may be considered eligible under the following two circumstances. In the case of a test pilot, a CAA test pilot, or an ASI, and in accordance with

GACAR § 100.225(b)(1), test pilots, CAA test pilots, and ASIs may receive the instrument training from an instructor pilot in an experimental powered-lift in accordance with the manufacturer's proposed training curriculum if the test pilot, CAA test pilot, or ASI is authorized to act as PIC of the powered-lift and the curriculum includes 15 hours of instrument training on the areas of operation listed in Section IV(d) of Appendix B to GACAR Part 141. In the case of an instructor pilot, the person must have provided the manufacturer's proposed training curriculum in the experimental powered-lift. In either case, the applicant must receive a logbook or training record endorsement. In the case of a test pilot, a CAA test pilot, or an ASI applicant, this endorsement must be from an instructor pilot documenting completion of the manufacturer's training curriculum. In the case of an instructor pilot, the endorsement must be from a management official within the manufacturer's organization certifying the instructor pilot provided the training in accordance with the manufacturer's proposed training curriculum. An example of the training record endorsement can be found in Appendix A, Example Endorsements.

8.1.3.2 GACAR §§ 100.225(b)(2) and 100.227(b)(2) authorize an applicant for an instrument rating in a powered-lift to receive 3 hours of instrument training from an instructor pilot in the experimental powered-lift in preparation for the practical test within the preceding 2 calendar months from the month of the test.

8.1.3.3 GACAR §§ 100.225(b)(4) and 100.227(b)(4) permit test pilots, CAA test pilots, ASIs, and instructor pilots who do not meet the cross-country aeronautical experience requirements specified in Appendix E, paragraph (c)(2)(ii) to GACAR Part 61 to be eligible for an instrument rating if they complete the cross-country requirements set forth by § 100.235. These requirements are detailed in Chapter 3, Alternate Cross-Country Requirements.

8.1.3.4 All flights recorded in the applicant's logbook will include the requirements listed in GACAR § 100.213(b). This experience will be presented to an evaluator or inspector as part of the applicant's verification of records for a practical test.

Table 8-1. Alternate Provisions for an Instrument-Powered-Lift Rating: Test Pilots and Instructor Pilots

CURRENT FLIGHT TIME REQUIREMENTS (Appendix E, paragraph (c) to GACAR Part 61)	ALTERNATE PROVISIONS FOR TEST PILOTS, CAA TEST PILOTS, OR ASIs*	ALTERNATE PROVISIONS FOR INSTRUCTOR PILOTS*
*To use the alternate provisions below, the flights must be conducted in an experimental powered-lift at the manufacturer, and the test pilots, CAA test pilots, ASIs, and instructor pilots must be authorized by the President to act as PIC of the experimental powered-lift. Refer to GACAR §§ 100.225(a) and 100.227(a).		
Cross-Country Time as PIC		
50 hours of cross-country time as PIC, of which 10 hours must have been in a powered-lift.	A test pilot, a CAA test pilot, or an ASI who is not rated in the powered-lift may log PIC flight time in the experimental powered-lift toward the 10-hour cross-country time requirement if: (1) The test pilot, CAA test pilot, or ASI is the sole manipulator of the controls; (2) The test pilot, CAA test pilot, or ASI is acting as PIC of the powered-lift; and (3) The flight is conducted for the purpose of R&D or showing compliance. Refer to § 100.225(c).	An instructor pilot who is not rated in the powered-lift may log PIC flight time in the experimental powered-lift toward the 10-hour cross-country time requirement if: (1) The instructor pilot is acting as PIC of the powered-lift, and (2) The flight is conducted for the purpose of crew training. Refer to § 100.227(c).
Instrument Time on the Areas of Operation Listed in Section IV(d) of Appendix B to GACAR Part 141		
40 hours of actual or simulated instrument time in the areas of operation listed in Section IV(d) of Appendix B to GACAR Part 141 of which 15 hours must have been received from an authorized instructor who holds an instrument-powered-lift rating, and the instrument time includes:	A test pilot, a CAA test pilot, or an ASI may receive 15 hours of instrument training on the areas of operation listed in Section IV(d) of Appendix B to GACAR Part 141 from an instructor pilot (rather than an authorized instructor) if: (1) The training is conducted in accordance with the manufacturer's proposed training curriculum, and (2) The test pilot, CAA test pilot, or ASI receives an endorsement from an instructor pilot certifying satisfactory completion of the curriculum. Refer to § 100.225(b)(1).	In lieu of receiving the 15 hours of instrument training on the areas of operation listed in Section IV(d) of Appendix B to GACAR Part 141, an instructor pilot may provide this instrument training to the test pilot in accordance with the manufacturer's proposed training curriculum. The instructor pilot must receive endorsement from a management official certifying that the instructor pilot provided the training. Refer to § 100.227(b)(1).
(1) 3 hours of instrument flight training from an authorized instructor in a powered-lift that is appropriate to the instrument-powered-lift	A test pilot, a CAA test pilot, or an ASI may accomplish the practical test preparation with an instructor pilot (rather than an authorized instructor).	An instructor pilot may accomplish the practical test preparation with another instructor pilot (rather than an authorized instructor).

CURRENT FLIGHT TIME REQUIREMENTS (Appendix E, paragraph (c) to GACAR Part 61)	ALTERNATE PROVISIONS FOR TEST PILOTS, CAA TEST PILOTS, OR ASIs*	ALTERNATE PROVISIONS FOR INSTRUCTOR PILOTS*
rating within 2 calendar months before the date of the practical test.	Refer to § 100.225(b)(2).	Refer to § 100.227(b)(2).
(2) Instrument flight training on cross-country flight procedures, including one cross-country flight in a powered-lift with an authorized instructor that is performed under instrument flight rules (IFR), when a flight plan has been filed with an air traffic control (ATC) facility, that involves: • A flight of 250 nautical miles (NM) along airways or by directed routing from an ATC facility; • An instrument approach at each aerodrome; and • Three different kinds of approaches with the use of navigation systems.	A test pilot, a CAA test pilot, or an ASI may complete the cross-country flight with an instructor pilot (rather than an authorized instructor). Instead of completing one cross-country flight of 250 NM, a test pilot, a CAA test pilot, or an ASI may complete two cross-country flights, each of which must involve a flight of 100 NM along airways or by directed routing from an ATC facility. Refer to §§ 100.225(b)(4) and 100.235(a).	An instructor pilot may complete the cross-country flight with another instructor pilot (rather than an authorized instructor). An instructor pilot may complete the same alternate cross-country flights as a test pilot. Refer to §§ 100.227(b)(4) and 100.235(a).

8.2 Alternate Provisions for Initial Cadre of Instructors.

8.2.1 Facilitation of Training. Once the manufacturer has completed type certification of their powered-lift, GACAR Part 100 facilitates manufacturers providing training for an initial cadre of instructors employed at specific certificate holders (CH). The manufacturer's approved training curriculum will cover all the required areas of operation, and the manufacturer's instructor pilots may deliver this training. This approach of facilitating the use of instructor pilots at the manufacturer to deliver initial certification training to the first powered-lift applicants enables those most familiar with the aircraft (i.e., the manufacturers) to provide safe and effective training and provides the manufacturer the means to ensure that there are a sufficient number of qualified pilots to operate their aircraft and train other pilots to operate their aircraft.

8.2.2 Authorization. In accordance with GACAR § 100.229(a), the initial cadre instructor applicant must be authorized by the President to serve as an initial check pilot, a chief instructor, an assistant chief instructor, or a Training Center Evaluator (TCE) for the purpose of initiating training in a powered-lift under an approved training program under GACAR Part 135, 141, or 142, as appropriate. The CH should submit a request to GACA. Upon authorization, these applicants will receive a temporary LOA from the President to receive this training. As set forth by GACAR § 100.229 and subsequently explained in the paragraphs below, the applicant must satisfactorily complete that training curriculum, and the training must be provided by the

manufacturer in a type-certificated powered-lift with a pilot who serves as an instructor pilot for that manufacturer.

8.2.2.1 *Initial Experience for Initial Cadre Instructors.* GACAR § 100.229 provides alternate experience requirements for the initial cadre instructors. These specific alternate requirements are:

- Cross-country time as PIC, and
- Instrument training on the areas of operation listed in Section IV(d) of Appendix B to GACAR Part 141.

8.2.2.2 *Alternate Logging Requirement.* GACAR § 100.229(c) authorizes an initial cadre instructor applicant who is not rated in the powered-lift to log PIC flight time in the powered-lift toward the 10-hour cross-country time requirement for flights that are conducted in accordance with a manufacturer's training curriculum if the applicant is:

- The sole manipulator of the controls with an instructor pilot on board, and
- Performing the duties of PIC.

8.2.2.3 *Instrument Training on the Areas of Operation Listed in Section IV(d) of Appendix B to GACAR Part 141.*

8.2.2.3.1 GACAR § 100.229(b)(1) authorizes an initial cadre instructor applicant to receive instrument training from an instructor pilot (rather than an authorized instructor) in a type-certificated powered-lift at a manufacturer in accordance with the manufacturer's training curriculum, if the curriculum covers the areas of operation listed in Section IV(d) of Appendix B to GACAR Part 141.

8.2.2.3.2 Initial cadre instructor applicants may be eligible for an instrument-powered-lift rating once they complete the manufacturer's proposed training curriculum for that aircraft and records documenting that completion are made available to an evaluator.

8.2.2.3.3 GACAR § 100.229(b)(2) permits the initial cadre instructor applicant to receive 3 hours of instrument training in a type-certificated powered-lift at a manufacturer in preparation for the practical test within the preceding 2 calendar months from the month of the test, if it is received from an instructor pilot employed by the manufacturer.

8.2.2.3.4 Pursuant to GACAR § 100.229(b)(4), an initial cadre instructor applicant who does not meet the cross-country aeronautical experience requirements specified in Appendix E, paragraph (c)(2)(ii) to GACAR Part 61 may be considered eligible for an instrument rating if they complete the cross-country requirements listed in GACAR § 100.235. These requirements are detailed in Chapter 3 of this AC.

8.2.2.3.5 The initial cadre instructor applicant must obtain a logbook or training record endorsement from the instructor pilot certifying that the person completed the cross-country flight, as specified in GACAR § 100.213.

8.2.2.3.6 Once the initial cadre instructor satisfactorily completes the manufacturer's training curriculum, they will receive a logbook or training record endorsement from the instructor pilot certifying that they satisfactorily completed the training in accordance with GACAR § 100.229(b)(1)(ii).

Table 8-2. Alternate Provisions for an Instrument-Powered-Lift Rating: Initial Cadre of Check Pilots, Chief Instructors, Assistant Chief Instructors, and Training Center Evaluators (Initial Cadre Instructors)

CURRENT FLIGHT TIME REQUIREMENTS (Appendix E, paragraph (c) to GACAR Part 61)	ALTERNATE PROVISIONS FOR INITIAL CADRE OF CHECK PILOTS, CHIEF INSTRUCTORS, ASSISTANT CHIEF INSTRUCTORS, AND TCEs* (initial cadre instructors, herein referred to as applicants)
*To use the alternate provisions below, the flights must be conducted by an instructor pilot in a type-certificated powered-lift at the manufacturer, and the applicant must be authorized by the President to serve as an initial check pilot, a chief instructor, an assistant chief instructor, or a TCE for the purpose of initiating training in a powered-lift under an approved training program under GACAR Part 135, 141, or 142, as appropriate. Refer to GACAR § 100.229(a).	
Cross-Country Time as PIC	
50 hours of cross-country time as PIC, of which 10 hours must have been in a powered-lift.	An applicant who is not rated in the powered-lift may log PIC flight time in the powered-lift toward the 10-hour cross-country time requirement for flights that are conducted in accordance with a manufacturer's training curriculum if the applicant is: (1) The sole manipulator of the controls, (2) Manipulating the controls of the powered-lift with an instructor pilot on board, and (3) Performing the duties of PIC. Refer to § 100.229(c).
Instrument Time on the Areas of Operation Listed in Section IV(d) of Appendix B to GACAR Part 141	
40 hours of actual or simulated instrument time in the areas of operation listed in Section IV(d) of Appendix B to GACAR Part 141 of which 15 hours must have been received from an authorized instructor who holds an instrument-powered-lift rating, and the instrument time includes:	An applicant may receive 15 hours of instrument training on the areas of operation listed in Section IV(d) of Appendix B to GACAR Part 141 from an instructor pilot (rather than an authorized instructor) if: (1) The training is conducted in accordance with the manufacturer's training curriculum, and (2) The applicant receives an endorsement from an instructor pilot certifying satisfactory completion of the curriculum. Refer to § 100.229(b)(1).
(1) 3 hours of instrument flight training from an authorized instructor in a powered-lift that is	An applicant may accomplish the practical test preparation from an instructor pilot (rather than an authorized instructor).

CURRENT FLIGHT TIME REQUIREMENTS (Appendix E, paragraph (c) to GACAR Part 61)	ALTERNATE PROVISIONS FOR INITIAL CADRE OF CHECK PILOTS, CHIEF INSTRUCTORS, ASSISTANT CHIEF INSTRUCTORS, AND TCEs* (initial cadre instructors, herein referred to as applicants)
appropriate to the instrument-powered-lift rating within 2 calendar months before the date of the practical test.	Refer to § 100.229(b)(2).
(2) Instrument flight training on cross-country flight procedures, including one cross-country flight in a powered-lift with an authorized instructor that is performed under IFR, when a flight plan has been filed with an ATC facility, that involves: • A flight of 250 NM along airways or by directed routing from an ATC facility; • An instrument approach at each aerodrome; and • Three different kinds of approaches with the use of navigation systems.	An applicant may complete the cross-country flight with an instructor pilot (rather than an authorized instructor). Instead of completing one cross-country flight of 250 NM, an applicant may complete two cross-country flights, each of which must involve a flight of 100 NM along airways or by directed routing from an ATC facility. Refer to §§ 100.229(b)(4) and 100.235(a).

8.3 Alternate Provisions for Pilots Trained in an Approved Training Program at the CH. Newly rated initial cadre instructors for an operator, a pilot school, or a training center (i.e., GACAR Part 135 check pilots, Part 141 chief instructors or assistant chief instructors, or Part 142 TCEs who have been trained and certified at the manufacturer) will be proficient in the proposed training curriculum under their respective part because of their involvement in developing the proposed training program. This involvement will enable them to be in the best position to provide training to pilots enrolled in their respective CH's training program under GACAR Part 135, 141, or 142. The initial cadre of flight instructors will become fully qualified to serve in their designated function and receive a permanent LOA from the President to subsequently conduct the training. Because these pilots will now be authorized instructors (i.e., meet the requirements for instructors under their respective part), the pilots who will receive training under an approved training program do not require the same breadth of relief that instructor pilots, test pilots, and the initial cadre instructors need. Rather, the only relief necessary for pilots at an approved training program is that of flight simulation training device (FSTD) credit, and the cross-country relief in § 100.235 extended to all applicants. Also, GACAR § 100.243 allows for initial training toward part 61 pilot certificates not normally accomplished at a GACAR Part 135 CH, which would require the trainer to hold a flight instructor certificate. See Chapter 12, Requirements to Serve as a Powered-Lift Pilot Under Part 135, for additional information.

8.3.1 *Use of Full Flight Simulator (FFS)*. As specified in GACAR § 100.231(c), an applicant may credit up to 4 hours obtained in a Level C or higher FFS toward the requirement to obtain 10 hours of cross-country time as PIC in a powered-lift if:

- The FFS represents the powered-lift category,

- The applicant was performing the duties of PIC, and
- The flight simulates a cross-country flight and includes the performance of instrument procedures under simulated instrument conditions.

8.3.2 *Instrument Training on the Areas of Operation Listed in Section IV(d) of Appendix B to GACAR Part 141.* Pursuant to GACAR § 100.231(b), applicants who do not meet the cross-country aeronautical experience requirements specified in Appendix E, paragraph (c)(2)(ii) to GACAR Part 61 may be considered eligible for an instrument rating if they complete the cross-country requirements listed in GACAR § 100.235, which may be completed with an authorized instructor. These requirements are detailed in Chapter 3 of this AC.

Table 8-3. Alternate Provisions for an Instrument-Powered-Lift Rating: Persons Receiving Training Under a Part 135, 141, or 142 Approved Training Program

CURRENT FLIGHT TIME REQUIREMENTS (Appendix E, paragraph (c) to GACAR Part 61)	ALTERNATE PROVISIONS FOR PERSONS RECEIVING TRAINING UNDER A PART 135, 141, OR 142 APPROVED TRAINING PROGRAM* (herein referred to as applicants)
*To use the alternate provisions below, the applicant must receive training at an approved training program under GACAR Part 135, 141, or 142 for the purpose of obtaining a powered-lift category rating.	
Cross-Country Time as PIC	
50 hours of cross-country time as PIC, of which 10 hours must have been in a powered-lift.	An applicant may credit up to 4 hours obtained in a Level C or higher FFS toward the requirement to obtain 10 hours of cross-country time as PIC in a powered-lift if: (1) The FFS represents the powered-lift category, (2) The applicant was performing the duties of PIC, (3) The flight simulates a cross-country flight and includes the performance of instrument procedures under simulated instrument conditions, and (4) The flights are conducted in accordance with an approved training program under GACAR Part 135, 141, or 142. Refer to § 100.231(c).
Instrument Time on the Areas of Operation Listed in Section IV(d) of Appendix B to GACAR Part 141	
40 hours of actual or simulated instrument time in the areas of operation listed in Section IV(d) of Appendix B to GACAR Part 141 of which 15 hours must have been received from an authorized instructor who holds an instrument-powered-lift rating, and the instrument time includes:	No alternative.
(1) 3 hours of instrument flight training from an authorized instructor in a powered-lift that is appropriate to the instrument-powered-lift rating within 2 calendar months before the date of the practical test.	No alternative.
(2) Instrument flight training on cross-country flight procedures, including one cross-country flight in a powered-lift with an authorized instructor that is	Instead of completing one cross-country flight of 250 NM, an applicant may complete two cross-country flights, each of which must involve a flight of

CURRENT FLIGHT TIME REQUIREMENTS (Appendix E, paragraph (c) to GACAR Part 61)	ALTERNATE PROVISIONS FOR PERSONS RECEIVING TRAINING UNDER A PART 135, 141, OR 142 APPROVED TRAINING PROGRAM* (herein referred to as applicants)
performed under IFR, when a flight plan has been filed with an ATC facility, that involves: • A flight of 250 NM along airways or by directed routing from an ATC facility; • An instrument approach at each aerodrome; and • Three different kinds of approaches with the use of navigation systems.	100 NM along airways or by directed routing from an ATC facility. Refer to §§ 100.231(b) and 100.235(a).

CHAPTER 9. ALTERNATE REQUIREMENTS FOR POWERED-LIFT WITHOUT FULLY FUNCTIONING DUAL CONTROLS USED IN FLIGHT TRAINING

9.1 Alternate Requirements for Flight Training Using an Aircraft Without Fully Functioning Dual Controls.

9.1.1 A person may apply for a powered-lift category rating, an instrument-powered-lift rating, and a powered-lift type rating concurrently for a powered-lift with single controls and a single pilot station under an approved GACAR Part 135, 141, or 142 training program by meeting the requirements listed in GACAR § 100.253(b) and GACAR Part 100 Appendix A.

9.1.2 Applicability of this provision applies to an approved GACAR Part 135, 141, or 142 training program for a powered-lift aircraft originally type certificated or seeking type certification with one set of controls and a single pilot station.

9.2 Required Training. An applicant must complete training in the order presented below.

9.2.1 *Ground Training.*

9.2.1.1 An applicant must receive and log the required ground training set forth by GACAR §§ 61.83(b) and Section III(b) of Appendix B to GACAR Part 141 prior to receiving any flight training.

9.2.1.2 Upon completion of the ground training, an applicant must successfully complete a written or oral check administered by the training provider.

9.2.2 *Full Flight Simulator (FFS) Training.* In lieu of flight training in an aircraft, section 4 of GACAR Part 100 Appendix A permits the flight training requirements referenced in GACAR §§ 100.217 through 100.235 (except for the alternate cross-country experience requirements listed in GACAR § 100.233), as well as any other applicable flight training requirements under GACAR Parts 61 and 100 to be accomplished in an FFS.

9.2.2.1 A training program would have to be submitted by the operator and approved by the President and must incorporate all the flight training necessary for an eligible applicant to receive a powered-lift category rating, an instrument-powered-lift rating, and type ratings under GACAR Parts 61 and 100. At a minimum, the training program must include at least 20 hours of flight training in an FFS for the commercial pilot powered-lift category rating and 15 hours of flight training in an FFS for the instrument-powered-lift rating.

9.2.2.2 Upon completion of the ground training requirements above, an applicant may complete all applicable flight training requirements prescribed in GACAR Part 61 or the corresponding framework prescribed in GACAR §§ 100.217 through 100.235 (except for the alternate cross-country requirements outlined in GACAR § 100.233), as applicable, in a qualified Level C or higher FFS representative of the powered-lift type with single controls and a single pilot station.

9.2.2.3 Upon completion of the simulator training, an applicant must satisfactorily accomplish a check in a qualified Level C or higher FFS in accordance with the approved training program.

9.2.2.3.1 This check initiates a 90-day window in which the applicant must complete a familiarity flight and demonstration flight outlined in the following paragraphs (e.g., if a check in the FFS was completed on January 2, 2026, the applicant must complete a familiarity flight and demonstration flight as outlined below before April 1, 2026).

9.2.3 Aircraft Familiarity Flight.

9.2.3.1 The purpose of this segment is to expose the applicant to the operation of the aircraft in actual flight while the person administering the familiarity flight demonstrates the maneuvers and procedures necessary for the ratings sought.

9.2.3.2 Within 90 days upon the satisfactory completion of the check above, an applicant must observe an aircraft familiarity flight in the powered-lift with an authorized instructor.

9.2.3.3 The aircraft familiarity flight must be accomplished in the aircraft with an instructor piloting the aircraft and the applicant having an unobstructed visual sight of the controls, an unobstructed visual sight of the instrumentation, and the ability to engage in active communication with the instructor during the entire flight.

9.2.3.4 The person authorized to conduct the flight must demonstrate all of the maneuvers and procedures required for the issuance of a Commercial Pilot Certificate with a powered-lift category rating, for the issuance of an instrument-powered-lift rating, and for the issuance of a powered-lift type rating.

9.2.3.5 Upon successful completion of the familiarity flight, the authorized instructor may endorse the applicant to act as pilot in command (PIC) of the specific powered-lift for the sole purpose of accomplishing the required demonstration flight with an authorized instructor on board. This demonstration flight is described below.

9.2.3.5.1 The expiration date of the endorsement specified above must not be longer than 90 days from the date of the successful completion of the check in the FFS.

9.2.4 Demonstration Flight.

9.2.4.1 The purpose of a demonstration flight is to enable an authorized instructor to ascertain that an applicant possesses satisfactory proficiency in the operation of the aircraft to then act as PIC safely in solo flight.

9.2.4.2 The demonstration flight requires the applicant to pilot the aircraft and demonstrate all the maneuvers and procedures required for the issuance of a Commercial Pilot Certificate with a powered-lift category rating, for the issuance of an instrument-powered-lift rating, and for the issuance of a powered-lift type rating while an authorized instructor observes the applicant.

9.2.4.3 The authorized instructor must have an unobstructed visual sight of the controls and instrumentation and maintain active communication with the applicant for the entire flight.

9.2.4.4 Upon successful completion of the demonstration flight and the applicant having demonstrated satisfactory proficiency in the aircraft, the check pilot, Training Center Evaluator (TCE), authorized instructor, instructor pilot, aviation safety inspector (ASI), or another person authorized by the President to administer this check may endorse the applicant to act as PIC of the specified powered-lift to conduct solo flight(s), subject to any conditions and limitations the instructor believes to be warranted.

9.2.4.4.1 The endorsement is valid for no more than 90 days from the date of successful completion of the demonstration flight.

9.2.5 *Failure to Complete Demonstration Flight During Specified Endorsement Period.* If an applicant does not successfully complete the demonstration flight within 90 days from the day of the FFS check, they may receive another endorsement if they complete the following requirements:

1. The applicant must return to the FFS and successfully complete a demonstration of proficiency, which must consist of the maneuvers listed in section 10 of GACAR Part 100 Appendix A, unless otherwise waived in accordance with GACAR § 100.207(c); and
2. The applicant must complete another aircraft demonstration flight in accordance with paragraph 9.2.4 above.

9.2.6 *Solo Flights.*

9.2.6.1 As a result of the applicant completing all of the required flight training in an FFS and not in the actual flight environment, the GACA increased the amount of solo flight time to 20 hours. This accounts for the amount of PIC flight time required by GACAR § 100.216(a). For example, the applicant will receive credit for 15 hours of FFS toward the PIC requirements outlined in GACAR § 100.216(a). The applicant would have 20 hours of PIC flight experience remaining. Because the remaining PIC experience will be gained as solo experience, the applicant will have met the required 35 hours of PIC flight experience listed in GACAR § 100.216(a).

9.2.6.2 An applicant has 90 days to complete the following solo flight training requirements:

1. Ten hours of solo flight time in accordance with Appendix D, paragraph (e)(4) to GACAR Part 61, or the applicant may utilize the corresponding alternate provision for the cross-country time as set forth in GACAR §§ 100.217 through 100.231, as applicable;
2. Ten hours of solo flight time that must include maneuvers and procedures required for the issuance of a Commercial Pilot Certificate with a powered-lift category rating, for the issuance of an instrument-powered-lift rating, and for the issuance of a powered-lift type rating; and

3. If the aircraft is unable to comply with the standard cross-country requirements then the applicant must also comply with the alternate cross-country aeronautical experience requirements prescribed in GACAR § 100.233.

9.2.7 Lapse of Endorsement for Solo Flight.

9.2.7.1 If an applicant is unable to complete the solo requirements within the 90-day endorsement window, then the applicant will need to meet specific requirements to receive another 90-day endorsement. There are two avenues an applicant can take to receive another endorsement. Below are the options to receive another solo endorsement.

9.2.7.1.1 *Re-endorsement within 90-day endorsement window.* If the applicant identifies that they will not complete the requirements within the timeframe, they will be required to complete another abbreviated demonstration flight with an authorized instructor. The requirements for the demonstration flight are found in section 10 of GACAR Part 100 Appendix A. Once the applicant successfully completes the demonstration flight, they may receive another 90-day endorsement.

9.2.7.1.2 *Re-endorsement outside of the 90-day endorsement window.* If an applicant does not complete the solo requirements and the 90-day endorsement has lapsed, the applicant will need to demonstrate proficiency in an FFS, which will result in an applicant being endorsed to accomplish another demonstration flight. The applicant must complete the demonstration flight in the powered-lift, and upon successfully completing the demonstration flight, the applicant may receive another 90-day endorsement to complete the solo requirements.

9.3 Practical Test. Upon completion of the requirements listed above and meeting the applicable regulatory requirements of GACAR Part 61 or 100, as applicable, the applicant may apply for a Commercial Pilot Certificate with a powered-lift category rating, an instrument-powered-lift rating, and a type rating in the specific powered-lift for which training under GACAR Part 100 Appendix A has occurred.

9.3.1 *Training Prerequisite.* In accordance with Appendix D, paragraph (e)(3)(iv) to GACAR Part 61, an applicant must complete 3 hours of training on the areas of operation listed in Appendix C, Section V, paragraph (d)(5) of GACAR Part 141 or the corresponding section in GACAR Part 100 in an FFS in preparation for the practical test within the preceding 2 calendar months from the month of the practical test. Section 7(a) of GACAR Part 100 Appendix A allows this requirement to be met using a Level C or higher FFS.

9.3.1.1 GACAR Part 100 allows for a person to complete the full training program in such a timeline that would allow the crediting of the FFS training in section 4 of GACAR Part 100 Appendix A to meet the requirements of Appendix D, paragraph (e)(3)(iv) to GACAR Part 61 and section 7(a) of GACAR Part 100 Appendix A. In other words, if the training program is completed within the preceding 2 calendar months, the flight training conducted during that period would be counted for purposes of both the 20 hours of flight training and the 3 hours of

practical test preparation. The applicant would perform a lookback as is normally accomplished when receiving other ratings today.

9.3.2 Practical Tests Conducted in an Aircraft. GACAR § 61.45(e) permits the use of an aircraft with a single set of controls for a practical test if the following provisions are met.

9.3.2.1 The evaluator must agree to conduct the test in the aircraft, and they must be located in a position to observe the proficiency of the applicant. The examiner is responsible for ensuring the practical test is in compliance with GACAR § 61.27, by encompassing all required tasks and maneuvers listed in the applicable Airman Certification Standards (ACS).

9.3.2.2 GACAR § 61.27(c)(5)(ii) does not allow for demonstration of instrument skills during a practical test with aircraft that have one set of flight controls. As a result, the instrument tasks are required to be completed in an FFS. If the practical test cannot be completed in its entirety in the aircraft, and some tasks are required to be accomplished in the FFS, then the satisfactory outcome of the practical test will result with the issuance of the Temporary Airman Certificate with the following limitations. The exception to this limitation is if the applicant meets the requirements of GACAR § 61.87(e).

9.3.2.2.1 The pilot will receive a limitation stating the certificate is subject to certain PIC limitations and will be restricted from the carriage of persons or property other than necessary for the conduct of the flight. Additionally, the carriage of property for compensation or hire is not permitted, but the carriage of property under GACAR Part 91 would be permitted, such as company materials. The limitations may be removed when the pilot meets the requirements listed below in paragraph 9.4.

9.4 Supervised Operating Experience (SOE) for a Powered-Lift with Single Controls and a Single Pilot Station. If a pilot receives a limitation as indicated above, they must meet the requirements set forth in section 8 of GACAR Part 100 Appendix A to have the limitation(s) removed. The following describes the requirements to get the limitation(s) removed.

9.4.1 Applicants Adding an Initial Powered-Lift Rating with Single Flight Controls and a Single Pilot Station. The applicant must perform 10 hours of flight time in an aircraft of category and type for which the limitation applies under the observation of a pilot who holds a category and type rating without limitations for the aircraft, maintaining full communication with the observing pilot.

9.4.1.1 The flight time must also include a cross-country flight in a powered-lift under instrument flight rules (IFR) on a flight plan filed with an air traffic control (ATC) facility that involves a flight of at least 100 nautical miles (NM) along airways or by directed routing from an ATC facility, an instrument approach at each aerodrome, and three different kinds of approaches with the use of navigation systems.

9.4.1.2 Since there is only a single set of controls and a single pilot station, the applicant must accomplish the cross-country flight detailed in paragraph 9.4.1.1 without the use of a view-limiting device.

9.4.1.3 The applicant must log each flight and the observing pilot attest in writing to each flight.

9.4.1.4 The applicant must present evidence of the SOE to any examiner or GACA to have the limitation removed.

9.4.2 Applicants Who Already Hold Powered-Lift Ratings and Are Only Adding an Additional Powered-Lift Type Rating in a Powered-Lift with Single Flight Controls and a Single Pilot Station. Applicants must comply with GACAR § 100.255 for an applicant adding an additional powered-lift type rating. Despite the requirements of GACAR § 61.87(f), a person who holds a powered-lift category rating, an instrument-powered-lift rating, and a powered-lift type rating and seeks an additional type rating for a powered-lift with single flight controls and a single pilot station in accordance with GACAR §§ 61.83(d) and 61.87(a), and does not meet requirements of GACAR § 61.87(e), will receive a limitation on the certificate restricting the person operating the powered-lift type from carrying any person or property on the aircraft, other than necessary for the purpose of removing the limitation from their certificate listed in GACAR § 100.255(b). The applicant must complete the following requirements.

9.4.2.1 The applicant must perform 25 hours of flight time in the type of powered-lift for which the limitation applies under the observation of a fully rated pilot without limitations for the aircraft, maintaining full communication with the observing pilot.

9.4.2.1.1 The observing pilot must have an unobstructed visual sight of the controls and instrumentation.

9.4.2.2 The applicant must log each flight and the observing pilot attest in writing to each flight.

9.4.2.3 The applicant must present evidence of the SOE to any examiner or GACA to have the limitation removed.

CHAPTER 10. MISCELLANEOUS QUALIFICATION REQUIREMENTS

10.1 Requirements for Pilots Serving as Second in Command (SIC) in a Powered-Lift.

10.1.1 *Application of GACAR § 61.19 to Powered-Lift Pilots.* Pilots seeking to serve as SIC of a powered-lift that is type certificated for more than one required pilot flight crewmember or in operations requiring an SIC pilot flight crewmember must meet the SIC qualification requirements of GACAR § 61.19. According to GACAR § 61.19(a), an individual must hold a powered-lift category rating on their pilot certificate as well as an instrument-powered-lift rating if the flight will be conducted under instrument flight rules (IFR). These requirements ensure that the SIC has obtained a base level of experience in the powered-lift category and successfully passed a practical test to obtain the powered-lift rating. With exception to circumstances described in paragraph 10.1.2 below, this experience may be in a different type of powered-lift than the powered-lift for which SIC privileges are sought.

10.1.2 *Airman Certification Standards (ACS) Task Requirements for Powered-Lift Type Rated Pilots.*

10.1.2.1 In accordance with GACAR § 100.209, in order for a pilot to act as SIC in a powered-lift, they must have completed all ACS tasks required for the practical test. There may be situations whereby a pilot obtains a powered-lift type rating but did not complete all the ACS tasks because the powered-lift was not capable of performing certain functions (as explained in Chapter 4, Alternate Requirements for Practical Tests and Training in a Powered-Lift). To ensure that the pilot is proficient in the maneuvers that the powered-lift is capable of, but that the pilot was not trained or tested on, the pilot must accomplish certain requirements to act as SIC in that aircraft that is capable of completing such tasks. GACAR § 100.209 sets forth the requirements an individual must accomplish to serve as SIC in a powered-lift in this unique circumstance. Therefore, in order to operate as SIC in that aircraft, that person must first have completed the following:

1. Received and logged ground and flight training from an authorized instructor on the specific tasks that were waived on the person's practical test; and
2. Received a logbook or training record endorsement from an authorized instructor certifying the person has satisfactorily demonstrated proficiency of those tasks. The authorized instructor should utilize the applicable ACS to determine whether a pilot has demonstrated proficiency of a task.

10.1.2.2 The training and endorsement are not required for those pilots who:

1. Satisfactorily complete a practical test in that aircraft, and that test includes each task required in GACAR § 61.27(b)(1)(i) (i.e., the tasks specified in the areas of operation

contained in FAA-S-ACS-17, Airline Transport Pilot and Type Rating for Powered-Lift Category Airman Certification Standards); or

2. Receive ground and flight training and satisfactorily complete a competency check under GACAR § 135.349 in a powered-lift, provided the training program and check include each task that was previously waived in accordance with GACAR § 100.207(b).

10.2 Training in a Powered-Lift that Requires Two Pilot Flight Crewmembers. An applicant for a powered-lift rating may receive training under GACAR §§ 100.221, 100.223, 100.229, and 100.231, notwithstanding the SIC requirements of GACAR § 61.19, in a powered-lift that is type certificated for more than one required pilot flight crewmember and may serve as SIC in a powered-lift that is type certificated for more than one required pilot flight crewmember without meeting the requirements of GACAR § 61.19(a)(1), (a)(2), and (b)(2).

10.3 Reserved.

10.4 Supervised Operating Experience (SOE) Requirements of GACAR § 61.87.

10.4.1 *Airmen Qualification Requirements for Type Rating Practical Test in a Full Flight Simulator (FFS).*

10.4.1.1 GACAR § 61.87(e) states that if a powered-lift is not used during the practical test for a type rating in a powered-lift (except for preflight inspection), an applicant must accomplish the entire practical test in a Level C or higher FFS and have 500 hours of flight time in the type of powered-lift for which the rating is sought. If the applicant meets the aeronautical experience requirements set forth in GACAR § 61.87(e), as appropriate to the type rating sought, then the applicant receives a type rating without limitation. An applicant may obtain the 500 hours of flight time that is required by GACAR § 61.87(e) while serving as SIC, as specified in GACAR § 100.209.

10.4.1.2 If the applicant does not satisfy the aeronautical experience requirements, then the applicant receives a PIC limitation on their pilot certificate in accordance with GACAR § 61.87(f)(2). The PIC limitation restricts the pilot from serving as PIC in the type of aircraft for which they have been type rated. The PIC limitation may be removed from the pilot certificate when the applicant complies with the SOE requirements set forth in GACAR § 61.64(g). Specifically, the applicant must perform 25 hours of flight time in an aircraft of the category, class, and type for which the limitation applies under the direct observation of a qualified PIC who holds the appropriate ratings, without limitations, for the aircraft. The applicant must obtain this SOE while performing the duties of PIC. Once the applicant completes the flight time required by GACAR § 61.64(g), they should contact GACA to have the limitation removed.

10.4.2 *Newly Rated Powered-Lift Pilots Utilizing SOE.* Before newly rated powered-lift pilots may perform SOE in a powered-lift, there first must be a cadre of qualified PICs to directly observe the flight time. Based on the experience test pilots gain in the specific type of powered-

lift, they will initially be the only pilots with enough flight time to qualify as PIC with a Commercial Pilot Certificate with a powered-lift category rating and a type rating without limitation. When these pilots are certificated, they will be able to directly observe SOE for applicants who receive a limitation on their certificate in accordance with GACAR § 61.87(f).

Note: SOE for GACAR Part 61 and SOE for GACAR Part 135 can be conducted concurrently as long as the aircraft and pilots meet the GACAR Part 135 regulatory requirements.

CHAPTER 11. RESERVED

CHAPTER 12. REQUIREMENTS TO SERVE AS A POWERED-LIFT PILOT UNDER PART 135

12.1 Alternate Provisions for Persons Receiving Training Under a Part 135 Approved Training Program.

12.1.1 *Pilot Eligibility.* As specified in GACAR § 100.243(a)(3), a GACAR Part 119 certificate holder (CH) authorized to conduct GACAR Part 135 operations may train a pilot for powered-lift ratings only if the pilot is employed by the part 119 CH and holds at least a Commercial Pilot Certificate with either:

- An airplane category rating with a single-engine or multiengine class rating and an instrument-airplane rating, or
- A rotorcraft category rating with a helicopter class rating and an instrument-helicopter rating.

12.1.2 *GACAR Part 61 Training and Endorsement Requirements.* An employee of a GACAR Part 135 CH may forgo the training and endorsement required in GACAR § 61.27 if the pilot receives a flight training record endorsement from the CH attesting that the person completed the CH's approved ground and flight training program. GACAR § 100.243(b) permits a pilot to apply to add a powered-lift category rating concurrently with an instrument-powered-lift rating and an initial powered-lift type rating to their Commercial Pilot Certificate, provided the person has a training record endorsement from an authorized instructor for that CH attesting that the pilot satisfactorily completed the applicable ground and flight training curricula in the approved GACAR Part 135 training program and the pilot was found proficient in the areas of operation listed in Appendix C, Section V, paragraph (d)(5) of GACAR Part 141 and Section IV(d) of Appendix B to GACAR Part 141 and GACAR § 61.179(e)(3).

12.1.3 *Checking and Testing.*

12.1.3.1 GACAR § 100.243(b) allows for a pilot who completes the certification curriculum and the part 135 CH training to apply to add a powered-lift category rating concurrently with an instrument-powered-lift rating and an initial powered-lift type rating to their Commercial Pilot Certificate if the person successfully completes the written or oral testing under GACAR § 135.349(a)(2) and (3); a competency check under GACAR § 135.349(b); and an instrument proficiency check (IPC) under GACAR § 135.351 provided certain conditions are met.

12.1.3.2 First, under GACAR § 100.243(b)(1)(iii)(A), the GACAR § 135.349(b) competency check is required to include the maneuvers and procedures required by part 61 for the issuance of a Commercial Pilot Certificate with a powered-lift category rating and a powered-lift type rating.

12.1.3.3 Second, under GACAR § 100.243(b)(1)(iii)(B), the IPC must meet the requirements of GACAR § 135.351, as applicable to a pilot in command (PIC) holding a Commercial Pilot Certificate, which includes the maneuvers and procedures required by GACAR Part 61 for the issuance of an instrument-powered-lift rating.

12.1.3.4 Finally, under GACAR § 100.243(c), the testing, competency check, and IPC will be administered by an aviation safety inspector (ASI), an Aircrew Program Designee (APD) who is authorized to perform competency checks and IPCs for the CH, or a Training Center Evaluator (TCE) with appropriate certification authority who is also authorized to perform competency checks and IPCs for the CH.

12.1.3.5 Additionally, GACAR § 135.355(b) allows a pilot who fails a maneuver on a check to be given additional training during the check and be retested while conducting the check. However, that is not authorized in under GACAR § 100.243(b)(2). This is because the checks performed result in the issuance of an additional rating in which similar provisions are not allowed for in pilot certification. GACAR § 100.243(b)(2) requires that if a pilot fails a maneuver on the competency check or IPC, the check and practical test result in an unsatisfactory outcome.

12.1.3.6 GACAR § 100.243(b)(2) also does not allow for substitution of an IPC for a competency check, as stated in GACAR § 135.349(c). Both the competency check and IPC must be completed.

12.1.4 *Alternate Qualification Requirements for Flight Instructors under Part 135.*

12.1.4.1 Normally, pilots hold the applicable category and class ratings prior to employment at a GACAR Part 135 CH. However, a GACAR Part 135 CH conducting powered-lift operations may not be able to hire pilots who hold the necessary powered-lift category ratings on their Commercial Pilot Certificates due to the aircraft being new to the civilian market. As a result, these pilots may receive their powered-lift category rating through an approved training program at the CH. The FAA has determined that, when complying with GACAR § 100.243, a person must hold a flight instructor certificate issued under part 61 with the appropriate ratings to provide training for the purpose of adding a powered-lift category rating or an instrument-powered-lift rating to a Commercial Pilot Certificate, pursuant to GACAR § 100.243(a)(2). This determination is based on:

- The lack of powered-lift experience held by pilots completing the GACAR Part 135 training program, and
- The curriculum content required for the issuance of a Commercial Pilot Certificate with a powered-lift category rating and an instrument-powered-lift rating.

12.1.4.2 Since GACAR § 100.203 requires all instructors to hold a flight instructor certificate with powered-lift ratings when providing instruction under an approved GACAR Part 135

training program for initial powered-lift ratings, a person who only holds an Airline Transport Pilot (ATP) Certificate with a powered-lift rating is unable to provide instruction for this specific training curricula in accordance with GACAR § 61.189(a).

12.2 Certification Training Curricula. GACAR § 100.243(a) authorizes a GACAR Part 119 CH authorized to conduct GACAR Part 135 operations to provide ground and flight training to meet the GACAR Part 61 requirements of a commercial pilot powered-lift rating, an instrument-powered-lift rating, and an aircraft type rating. The approved training curricula must include all applicable requirements in GACAR Parts 61, 135, and 194 for the addition of a powered-lift category rating to a Commercial Pilot Certificate and instrument-powered-lift rating.

12.3 Reserved.

12.4 PIC Qualifications. GACAR § 135.341 prescribes qualifications for pilots serving as PIC, which also includes pilots serving as PIC in powered-lift. Table 12-1 below indicates the minimum level of certificate and ratings the PIC is required to hold for certain operations conducted under GACAR Part 135.

Table 12-1. Powered-Lift Pilot-in-Command Qualifications

Part 135 Operation	Minimum Certificate Required	Minimum Ratings Required
[Reserved]	[Reserved]	[Reserved]
Non-turbojet-powered operations in powered-lift with nine or less passenger seats, excluding crewmember seats	Commercial certificate	1. Powered-lift category rating; 2. Appropriate powered-lift type rating, not limited to VFR; and 3. Instrument-powered-lift rating or an ATP Certificate not limited to VFR with a powered-lift category rating
Cargo operations	Commercial certificate	1. Powered-lift category rating; 2. Appropriate powered-lift type rating, not limited to VFR; and 3. Instrument-powered-lift rating or an ATP Certificate not limited to VFR with a powered-lift category rating.

12.5 Reserved.

CHAPTER 13. REQUIREMENTS FOR PART 141 SCHOOLS

13.1 Class Rating Requirements for Part 141. Under GACAR Part 141, flight training requirements include powered-lift category and class ratings. However, due to the inability to establish classes for powered-lift ratings at this time, GACAR Part 100 exempts a GACAR Part 141 certificate holder (CH) from requiring a class rating in an approved course.

13.2 Alternate Qualification Requirements for Chief Instructors, Assistant Chief Instructors, and Check Instructors. As specified in GACAR § 100.241, relief is provided to the requirements of GACAR §§ 141.51(a)(1), 141.57(a)(1), and 141.59(a)(2)(ii) that a person hold a powered-lift class rating on their pilot and flight instructor certificates. However, GACAR § 100.241 requires that each person seeking designation as a chief instructor, an assistant chief instructor, or a check instructor for checks and tests that uses a powered-lift holds the specific type rating used in that course of training. Specifically, any person seeking to qualify as a chief instructor, an assistant chief instructor, and a check instructor must meet the following requirements:

1. Hold a Commercial Pilot Certificate or an Airline Transport Pilot (ATP) Certificate with the following ratings:
 - a. A powered-lift category rating;
 - b. A type rating for the powered-lift used in the course; and
 - c. An instrument-powered-lift rating, if an instrument rating is required for the course.
2. Hold a current flight instructor certificate with the following ratings:
 - a. A powered-lift category rating; and
 - b. An instrument-powered-lift rating, if an instrument rating is required for the course.

13.3 Alternate Qualification Requirements for Check Instructors for Checks and Tests That Relate to Ground Training. As specified in GACAR § 100.241(b), a person seeking designation as a check instructor for checks and tests that relate to ground training for powered-lift must hold a current flight instructor certificate with a powered-lift category rating or ground instructor certificate with ratings appropriate to the category of aircraft used in the course.

13.4 Alternate Means to Satisfy Flight Training Minimum Curriculum Content Specified in Part 141 Appendices for a Powered-Lift Category Rating. Pilot schools seeking to utilize a powered-lift in a course of training are required to comply with appendices to GACAR Part 141 for the respective course they are offering. Powered-lift courses are outlined in these appendices and reference the minimum areas of operation to train applicants for the certificate they are seeking. Under GACAR § 100.207, the GACA is allowing for a task to be waived when it is deemed the powered-lift is not capable of the task through the aircraft certification process. Since GACAR Part 141 requires training and testing to mirror GACAR Part 61, a pilot school might also wish to utilize this flexibility when applying for a course of training with a powered-lift. Therefore, the GACA is also allowing a GACAR Part 141 pilot school seeking approval of a course in a powered-lift resulting in a Private or Commercial Pilot Certificate to comply with GACAR § 100.239(a) in lieu of the minimum curriculum content when the powered-lift they seek to operate is not deemed capable of a certain task through the aircraft certification process.

13.5 Alternate Cross-Country Requirements Applicable to Part 141 Approved Courses. Powered-lift courses are outlined in the appendices of GACAR Part 141 and reference the cross-country requirements applicants must complete for the certificate they are seeking. Because the cross-country requirements mirror those in GACAR Part 61 the GACA is allowing for a GACAR Part 141 course to use those cross-country requirements under the parameters specified in GACAR § 100.239(b) for the appropriate course by providing an alternate means to satisfy certain minimum curriculum content in GACAR Part 141 Appendices A, B, C, and L.

CHAPTER 14. PART 91 PILOT CERTIFICATION FOR POWERED-LIFT WITH SPECIAL AIRWORTHINESS CERTIFICATES FOR EXPERIMENTAL PURPOSES

14.1 Operating Limitations. Under GACAR § 21.173, an aircraft may be issued an Experimental Airworthiness Certificate for specific purposes. In general, experimental aircraft are not subject to the same airworthiness standards as their counterparts holding Standard Airworthiness Certificates. As a result of these differing standards, experimental aircraft are subject to additional operating limitations. Issuance of additional operating limitations is authorized by GACAR § 91.421(f), and compliance with these operating limitations is required by GACAR § 91.13(a). As such, category ratings and other authorizations (e.g., the authorization to act as pilot in command (PIC)) may be prescribed through operating limitations for experimental powered-lift, as warranted.

APPENDIX A. EXAMPLE ENDORSEMENTS

A.1 Cross-Country Endorsements for Each Certificate Type for All Applicants.

I certify that [First name, MI, Last name], [grade of pilot certificate], [certificate number], has completed the cross-country flights as specified in GACAR § 100.211(b)(1), (b)(2), (b)(3), and (c). I have determined that [First name, MI, Last name] meets the minimum cross-country aeronautical experience required for the commercial pilot powered-lift practical test.

Signature of Instructor _____ Print Name _____ Cert No. _____

Date _____

I certify that [First name, MI, Last name], [grade of pilot certificate], [certificate number], has completed the cross-country flights as specified in GACAR § 100.209(d)(1). I have determined that [First name, MI, Last name] meets the minimum cross-country aeronautical experience required for the instrument-powered-lift practical test.

Signature of Instructor _____ Print Name _____ Cert No. _____

Date _____

I certify that [First name, MI, Last name], [grade of pilot certificate], [certificate number], has completed the cross-country flights as specified in GACAR § 100.214(a), (b)(1), and (b)(2). I have determined that [First name, MI, Last name] meets the minimum cross-country aeronautical experience required for the private pilot powered-lift practical test.

Signature of Instructor _____ Print Name _____ Cert No. _____

Date _____

A.2 Test Pilot or Aviation Safety Inspector (ASI)—Commercial Pilot.

[Cross-country endorsement above]

I certify that [First name, MI, Last name], [grade of pilot certificate], [certificate number], has satisfactorily completed the training curriculum for a(n) [specify the manufacturer name and model of aircraft] as specified in GACAR § 100.217(b)(1).

Signature of Instructor Pilot _____ Print Name _____ Cert No. _____

Date _____

I certify that [First name, MI, Last name], [grade of pilot certificate], [certificate number], has received the required training in a(n) [specify the manufacturer name and model of aircraft] as specified in GACAR § 100.217(b)(1) and meets the aeronautical experience requirements of GACAR § 100.211(a)(1)(v)(A). I have determined that [First name, MI, Last name] is prepared for the required practical test for the issuance of a commercial pilot powered-lift certificate.

Signature of Instructor Pilot _____ Print Name _____ Cert No. _____

Date _____

I certify that [First name, MI, Last name], [grade of pilot certificate], [certificate number], has received and logged 3 hours of training time in the experimental powered-lift within 2 calendar months preceding the month of application in preparation for the commercial pilot practical test as specified in GACAR § 100.211(d)(1)(iii).

Signature of Instructor Pilot _____ Print Name _____ Cert No. _____

Date _____

A.3 Test Pilot or ASI—Instrument Rating.

[Cross-country endorsement above]

I certify that [First name, MI, Last name], [grade of pilot certificate], [certificate number], has satisfactorily completed the training curriculum for a(n) [specify the manufacturer name and model of aircraft] as specified in GACAR § 100.225.

Signature of Instructor Pilot _____ Print Name _____ Cert No. _____

Date _____

I certify that [First name, MI, Last name], [grade of pilot certificate], [certificate number], has received the required training in a(n) [specify the manufacturer name and model of aircraft] as specified in GACAR § 100.209(a)(1)(iii) and meets the aeronautical experience requirements of GACAR § 100.209(a)(1)(v)(A). I have determined that [First name, MI, Last name] is prepared for the required practical test for the issuance of an instrument-powered-lift rating.

Signature of Instructor Pilot _____ Print Name _____ Cert No. _____

Date _____

I certify that [First name, MI, Last name], [grade of pilot certificate], [certificate number], has received and logged 3 hours of training time in the experimental powered-lift within 2 calendar months preceding the month of application in preparation for the instrument rating practical test as specified in § GACAR 100.209(b)(1)(iii).

Signature of Instructor Pilot _____ Print Name _____ Cert No. _____

Date _____

A.4 Instructor Pilot—Commercial Pilot.

[Cross-country endorsement above]

I certify that [First name, MI, Last name], [grade of pilot certificate], [certificate number], has satisfactorily completed the training curriculum for a(n) [specify the manufacturer name and model of aircraft] as specified in GACAR § 100.219(b)(1).

Signature of Management Official _____ Print Name _____

Date _____

I certify that [First name, MI, Last name], [grade of pilot certificate], [certificate number], has completed the cross-country flights as specified in GACAR § 100.211(b)(1), (b)(2), and (b)(3) at [name of manufacturer] in a(n) [specify model of aircraft]. I have determined that [First name, MI, Last name] meets the minimum cross-country aeronautical experience required for the commercial pilot powered-lift practical test.

Signature of Management Official _____ Print Name _____

Date _____

I certify that [First name, MI, Last name], [grade of pilot certificate], [certificate number], has provided the required training in a(n) [specify the manufacturer name and model of aircraft] as specified in GACAR § 100.211(a)(1)(iii) and meets the aeronautical experience requirements of GACAR § 100.211(a)(1)(v)(B). I have determined that [First name, MI, Last name] is prepared for the required practical test for the issuance of a commercial pilot powered-lift certificate.

Signature of Management Official _____ Print Name _____

Date _____

I certify that [First name, MI, Last name], [grade of pilot certificate], [certificate number], has provided instruction and logged 3 hours of training time in the experimental powered-lift within 2 calendar months preceding the month of application in preparation for the commercial pilot practical test as specified in GACAR § 100.211(d)(1)(iii).

Signature of Management Official _____ Print Name _____

Date _____

A.5 Instructor Pilot—Instrument Rating.

[Cross-country endorsement above]

I certify that [First name, MI, Last name], [grade of pilot certificate], [certificate number], has satisfactorily completed the training curriculum for a(n) [specify the manufacturer name and model of aircraft] as specified in GACAR § 100.227.

Signature of Management Official _____ Print Name _____

Date _____

I certify that [First name, MI, Last name], [grade of pilot certificate], [certificate number], has completed the cross-country flights as specified in GACAR § 100.211(c)(1) at [name of manufacturer] in a(n) [specify model of aircraft]. I have determined that [First name, MI, Last name] meets the minimum cross-country aeronautical experience required for the instrument rating powered-lift practical test.

Signature of Management Official _____ Print Name _____

Date _____

I certify that [First name, MI, Last name], [grade of pilot certificate], [certificate number], has provided the required training in a(n) [specify the manufacturer name and model of aircraft] as specified in GACAR § 100.209(a)(1)(iii) and meets the aeronautical experience requirements of GACAR § 100.209(a)(1)(v)(B). I have determined that [First name, MI, Last name] is prepared for the required practical test for the issuance of an instrument-powered-lift rating.

Signature of Management Official _____ Print Name _____

Date _____

I certify that [First name, MI, Last name], [grade of pilot certificate], [certificate number], has provided instruction and logged 3 hours of training time in the experimental powered-lift within 2 calendar months preceding the month of application in preparation for the instrument rating practical test as specified in GACAR § 100.209(b)(1)(iii).

Signature of Management Official _____ Print Name _____

Date _____

A.6 Initial Cadre Pilots for Certificate Holders (CH).

A.6.1 Commercial Pilot.

[Cross-country endorsement above]

I certify that [First name, MI, Last name], [grade of pilot certificate], [certificate number], has satisfactorily completed the training curriculum for a(n) [specify the manufacturer name and model of aircraft] as specified in GACAR § 100.217(b)(1).

Signature of Instructor Pilot _____ Print Name _____ Cert No. _____

Date _____

I certify that [First name, MI, Last name], [grade of pilot certificate], [certificate number], has received the required training as specified in GACAR § 100.211(a)(2)(ii) and meets the aeronautical experience requirements of GACAR § 100.211(a)(2)(iii). I have determined that [First name, MI, Last name] is prepared for the required practical test for the issuance of a commercial pilot powered-lift certificate.

Signature of Instructor Pilot _____ Print Name _____ Cert No. _____

Date _____

I certify that [First name, MI, Last name], [grade of pilot certificate], [certificate number], has received and logged 3 hours of training time in a(n) [specify make and model] powered-lift within 2 calendar months preceding the month of application in preparation for the commercial pilot practical test as specified in GACAR § 100.211(d)(2)(ii).

Signature of Instructor Pilot _____ Print Name _____ Cert No. _____

Date _____

A.6.2 Instrument Pilot.

[Cross-country endorsement above]

I certify that [First name, MI, Last name], [grade of pilot certificate], [certificate number], has satisfactorily completed the training curriculum for a(n) [specify the manufacturer name and model of aircraft] as specified in GACAR § 100.225.

Signature of Instructor Pilot _____ Print Name _____ Cert No. _____

Date _____

I certify that [First name, MI, Last name], [grade of pilot certificate], [certificate number], has received the required training as specified in GACAR § 100.209(a)(2)(ii) and meets the aeronautical experience requirements of GACAR § 100.209(a)(2)(iii). I have determined that [First name, MI, Last name] is prepared for the required practical test for the issuance of an instrument-powered-lift rating.

Signature of Instructor Pilot _____ Print Name _____ Cert No. _____

Date _____

I certify that [First name, MI, Last name], [grade of pilot certificate], [certificate number], has received and logged 3 hours of training time in a(n) [specify make and model] powered-lift within 2 calendar months preceding the month of application in preparation for the instrument rating practical test as specified in GACAR § 100.209(b)(2)(ii).

Signature of Instructor Pilot _____ Print Name _____ Cert No. _____

Date _____

A.7 Training at the CH. In addition to the required GACAR Part 61 endorsements, applicants will need the following additional endorsements.

A.7.1 Commercial Pilot.

[Cross-country endorsement above]

I certify that [First name, MI, Last name], [grade of pilot certificate], [certificate number], has received the required training in accordance with [name of operator] approved [name of training program] and is prepared for the required practical test for the issuance of a commercial pilot powered-lift certificate.

Signature of Authorized Instructor _____ Print Name _____

Cert No. _____ Date _____

A.7.2 Instrument Pilot.

[Cross-country endorsement above]

I certify that [First name, MI, Last name], [grade of pilot certificate], [certificate number], has received the required training in accordance with [name of CH] approved [name of training curriculum] and is prepared for the required practical test for the issuance of an instrument-powered-lift rating.

Signature of Authorized Instructor _____ Print Name _____

Cert No. _____ Date _____

A.8 Single Flight Controls and Single Pilot Station.

A.8.1 Endorsement Given at the Completion of Familiarity Flight in Accordance with Section 5 of GACAR Part 100 Appendix A.

I certify that [First name, MI, Last name] has received the training as required by section 4 of GACAR Part 100 appendix A. I certify that they have completed a check in accordance with section 4(b) of GACAR Part 100 Appendix A on [Date of full flight simulator (FFS) check]. [First name, MI, Last name] is able to serve as a pilot in command (PIC) to perform a Demonstration Flight as required by section 5(d) of GACAR Part 100 Appendix A in a [Type of powered-lift]. This endorsement expires [90 days from the completion of FFS check indicated above].

A.8.2 Ninety-Day Solo Endorsement After Demonstration Flight (Initial and Re-Endorsement After Lapse).

I certify that [First name, MI, Last name] has completed the Demonstration Flight required by section 6(a)(4) of GACAR Part 100 Appendix A, which occurred on [Date of demonstration flight]. I certify that [First name, MI, Last name] can serve as a pilot in command (PIC) in a [Type of powered-lift] for the purposes of solo flights.

Limitations: [optional].

This endorsement expires [90 days from the completion of the Demonstration Flight indicated above].

A.8.3 Lapse in Solo 90-Day Endorsement.

I certify that [First name, MI, Last name] has completed the Demonstration of Proficiency as required by section 6(d)(2)(i) of GACAR Part 100 Appendix A on [Date applicant completed FFS demonstration of proficiency] and may serve as a pilot in command (PIC) to perform a Demonstration Flight as required by section 6(a)(4) of GACAR Part 100 Appendix A.

This endorsement expires [90 days from the date the applicant completed FFS demonstration of proficiency].

A.9 For all other endorsements, refer to FAA AC 61-65, Certification: Pilots and Flight and Ground Instructors (e.g., Solo, Second in Command (SIC), and Flight Instructor).

APPENDIX B. LIST OF ADVISORY CIRCULARS AFFECTED BY GACAR PART 100

B.1 This appendix lists the ACs (current editions) that are affected by GACAR Part 100. This list is not all-inclusive; it is the responsibility of the operator to ensure they are complying with all requirements when developing a training curriculum.

1. Reserved.
2. FAA AC 61-65, Certification: Pilots and Flight and Ground Instructors.
3. FAA AC 61-98, Currency Requirements and Guidance for the Flight Review and Instrument Proficiency Check.
4. FAA AC 61-136, FAA Approval of Aviation Training Devices and Their Use for Training and Experience.
5. Reserved.
6. Reserved.
7. Reserved.
8. Reserved.
9. FAA AC 141-1, Part 141 Pilot Schools, Application, Certification, and Compliance.
10. FAA AC 142-1, Standardized Curricula Delivered by Part 142 Training Centers.
11. GACA AC 194-01, Powered-Lift (VTOL) Operations.